Forcing global Π^2_n uniformization, with and without projective determinacy

Stefan Hoffelner*

TU Wien
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Abstract

We construct a set-forcing extension of L which satisfies the Π^2_n-uniformization property for every finite $n \geq 1$. Assuming ω Woodin cardinals and a measurable cardinal above them, we also obtain an extension of $L(\mathbb{R})^{M_\omega}$ which adds no reals and satisfies projective determinacy together with Π^2_n uniformization for every finite $n \geq 1$.

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1 Introduction

A *uniformization* of $A \subseteq \omega^\omega \times \omega^\omega$ is a function $U \subseteq A$ with $\operatorname{dom} U = \operatorname{dom} A$. A pointclass has the uniformization property if each of its relations has a uniformization in the pointclass. We work with second-order pointclasses over HC: first-order quantifiers range over HC, class quantifiers over all subsets of HC.

Definable uniformizations classically come from scales and determinacy; see Moschovakis [16]. At the second-order level the two sides behave differently. Under $\mathsf{AD} + V = L(\mathbb{R})$, the scale analysis of Steel [18] and Martin and Steel [14] gives the scale property, and hence uniformization, for Σ^2_1. Woodin extended the positive scale result to $\mathsf{ZF} + \mathsf{AD}^+ + V = L(\mathcal{P}(\mathbb{R}))$; see Steel and Trang [23, Theorem 2]. A Kechris–Solovay argument, akin to the classical argument by A. Levy, gives, under $\mathsf{AD} + V = L(\mathbb{R})$, a Π^2_1 relation with no uniformization in $L(\mathbb{R})$ at all [20, Theorem 3.11 and the following discussion].

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Positive results on the Π side come from games of length ω_1. Building on [21], Steel [22] showed that ω_1-open-projective determinacy gives the scale property for the open game quantifier on analytical trees: the games have length ω_1 and integer moves, and the trees of countable positions are definable over $(\mathrm{HC}, \in)$ from parameters. Under CH this pointclass is Π^2_1, so the hypothesis yields Π^2_1 uniformization [22, Corollary 0.6 and Theorem 2.1].

The relevant determinacy hypothesis follows from Neeman's work [17]: in the form quoted by Steel [22, Theorem 0.1], it suffices that every real belongs to a countable, $(\omega_1 + 1)$-iterable mouse satisfying $\mathsf{ZFC}^- +$ "there is a measurable Woodin cardinal." Steel distinguishes this mouse-existence hypothesis from an ordinary large-cardinal assumption and notes the unresolved iterability needed to derive it from the latter [22, footnote 1]. Neeman also records Steel's result that definable open length-ω_1 determinacy fails in the minimal iterable model with a class of indiscernible Woodin cardinals [17, p. 83].

A forcing antecedent is Abraham and Shelah's construction [1]: without adding reals, they obtain a Δ^2_2 wellorder of the reals and, by the same method, code prescribed relations on the reals. This gives an earlier instance of changing second-order definability while fixing the reals. Uniformization for a whole pointclass requires a definable choice for every relation in that pointclass as computed in the final model.

We obtain the Π side by forcing over L, at every finite level simultaneously.

Theorem 1.1. *There is a cardinal-preserving set-forcing extension $L[G]$ of L which adds no reals and satisfies Π^2_n-UP for every $n \geq 1$. The uniformizer uses the same real parameters as the relation; in particular, a lightface relation has a lightface uniformizer.*

The pointclasses are computed in the full extension. Since the forcing adds no reals and preserves ω_1, HC is unchanged, but the new subsets of HC change the meaning of class quantifiers. The theorem therefore concerns second-order definability in $L[G]$, whereas the determinacy results above concern their own ambient models. No large cardinals are used.

The construction is also compatible with projective determinacy.

Theorem 1.2. *Assume there are ω Woodin cardinals and a measurable cardinal above them, and put $\mathcal{W} = L(\mathbb{R})^{M_\omega}$. There is a forcing $\mathbb{P} \in \mathcal{W}$ which preserves all cardinals of $\mathcal{W}$, adds no reals, and forces $\mathsf{PD} + \mathsf{CH}$ and Π^2_n-UP with the same real parameters for every finite $n \geq 1$.*

The ground model $\mathcal{W}$ is its own $L(\mathbb{R})$ and satisfies $\mathsf{ZFC} + \mathsf{CH} + \mathsf{PD}$. Forcing without adding reals leaves $L(\mathbb{R})$, the projective payoffs and their strategy codes unchanged, and hence preserves PD. As before, only the new classes over HC change the second-order pointclasses.

Both ground models have the opposite pattern. Their canonical wellorders of the subsets of ω_1 are Σ^2_1-good (in $\mathcal{W}$ via a recoverable predicate A), and least coded witnesses give Σ^2_n uniformization for every finite $n \geq 1$. Since the two dual properties cannot hold at the same level, we obtain the following.

Corollary 1.3. *Under the hypothesis of Theorem 1.2, the models $\mathcal{W} \subseteq \mathcal{W}[G]$ both satisfy $\mathsf{ZFC} + \mathsf{PD} + \mathsf{CH}$, have the same* HC*, and satisfy, for every finite $n \geq 1$,*

$$\mathcal{W} \models \Sigma^2_n\text{-UP} \ \wedge \ \neg\mathbf{\Pi}^2_n\text{-UP},$$
$$\mathcal{W}[G] \models \Pi^2_n\text{-UP} \ \wedge \ \neg\mathbf{\Sigma}^2_n\text{-UP}.$$

The positive assertions hold with the same real parameters. The analogous contrast holds between L and the extension $L[G]$ of Theorem 1.1.

Thus, relative to the stated large-cardinal hypothesis, both full second-order uniformization patterns are consistent with $\mathsf{PD} + \mathsf{CH}$. The forcing turns one into the other while preserving the reals, HC and all projective truth; only the collection of classes over HC changes.

The good orders used for the ground-model Σ pattern are related to those studied by Lücke, Schindler and Schlicht [13]. Their Theorem 5.2 gives a good $\Sigma_1(\omega_1)$-definable wellorder of H_{ω_2} in M_1. They also prove that infinitely many Woodin cardinals and a measurable above them preclude a $\Sigma_1(\omega_1)$-definable uniformization of the club filter on ω_1 [13, Theorem 4.19]; there the inputs and outputs are subsets of ω_1, rather than reals.

Lücke and Müller [12, Theorem 1.6] show that the same large-cardinal hypothesis, or Woodin's axiom $(*)$, precludes a Σ_1-definable wellorder of any subset of $\mathcal{P}(\omega_1)$ of size greater than $\aleph_1$, with parameters in $\mathrm{HC} \cup \{\omega_1\}$. At the second-order level, Woodin's theorem rules out a Σ^2_1 wellorder of the reals under CH and a measurable Woodin cardinal [1, p. 2].

The method. Fix a Π^2_n formula $A_{n,e}(p, x, y)$ with formula number e and real parameter p. For fixed x, label each candidate y by the real $\ell(e, p, x, y)$, where ℓ is a fixed recursive injection. We define the uniformizer in the final extension by

$$U_{n,e}(p, x, y) \iff A_{n,e}(p, x, y) \wedge \neg\,\mathrm{Code}_n(\ell(e, p, x, y)).$$

Each real label t indexes a member S^n_t of a canonical family of stationary subsets of ω_n in L. Given a sequence $\vec{T} = (T_n)_{n\geq 1}$ of sets of labels, we construct a forcing $F_{\vec{T}}$ in whose extension

$$\mathrm{Code}_n(t) \iff S^n_t \text{ is nonstationary in } \omega_n \iff t \in T_n.$$

At level one, $\mathrm{Code}_1(t)$ simply asks for a club in ω_1 disjoint from S^1_t. At higher levels, a club in ω_n is too large to serve directly as a class over

HC. Here $\mathrm{Code}_n(t)$ asks for a set $X \subseteq \omega_1$ from which successive almost disjoint decoding gives a set $C_X \subseteq \omega_n$.[1] Every suitable coded model N must see $C_X \cap \omega_n^N$ as a club in ω_n^N disjoint from $S_t^n \cap \omega_n^N$. These ordinals ω_n^N are cofinal in ω_n. Since the models read initial segments of the same C_X, a witness yields a club disjoint from S_t^n in the final model. Section 7 expresses this test by a parameter-free Σ_n^2 formula. Thus $U_{n,e}$ is Π_n^2 with the original parameter p, and we must leave exactly one label uncoded in each nonempty section.

The difficulty is that the section itself depends on which labels we code. The forcing adds no reals, so every candidate y is already in L, but new subsets of HC can change the truth of $A_{n,e}(p, x, y)$. To determine in L which values a proposed $F_{\vec{T}}$ will force into a section, we reduce Π_n^2 truth in the full extension to $\Pi_1^2(H_{\omega_n})$ truth in the intermediate club extension. Cone homogeneity of the club forcing then gives a decision for each instance with ground-model real parameters.

We begin with every label in every T_n and treat the levels in increasing order. At level n, take an unassigned section for which $F_{\vec{T}} \Vdash A_{n,e}(p, x, y)$ for some y. Choose one such y and delete only $\ell(e, p, x, y)$ from T_n. Shrinking T_n with the other levels fixed preserves every Π_n^2 instance already forced, so y remains in its section. A previously empty section can become nonempty, so we repeat the choice, intersecting the remaining sets of labels at limits, until no unassigned nonempty section remains. Each deletion assigns a new section, so the recursion terminates before ω_2 by CH. Once $T_1, \dots, T_n$ are fixed, even simultaneous changes at all higher levels leave every level-n decision unchanged. Thus each nonempty section in the final model contains its assigned value, and its label is the only one omitted from coding in that section.

For Theorem 1.2 we write $\mathcal{W} = L[A]$, where $A \subseteq \omega_1^{\mathcal{W}}$ codes the natural enumeration of the reals, and relativize the construction to $(\mathrm{HC}, \in, A)$. Membership and nonmembership in A have parameter-free Σ_1^2 definitions which remain correct in all relevant extensions; they eliminate A from the uniformizers without raising complexity.

Organization. Section 2 collects preliminaries. Section 3 constructs the stationary sets and the club forcing, and Subsection 3.7 previews the case $n = 2$ as a warm up. It introduces one of the main ideas in that we work from outer models to inner models via deleting codes, even though simulating this process from the ground model. Sections 4 and 5 define the coding forcing and prove its structural and bookkeeping properties. Sections 6 and 7 give the reduction of complexity and the coding predicates. Sections 8 and 9

[1] For stationary-set coding see Friedman and Magidor [5], and for localization see Fischer and Friedman [3] and Fischer, Friedman and Zdomskyy [4]. A related application is given in [8].

compare the supports and complete the proof of Theorem 1.1. Section 10 relativizes the construction to $L[A]$, proves Theorem 1.2 and derives Corollary 1.3.

2 Preliminaries

2.1 Second-order pointclasses

Let H be a transitive set. A formula is $\Sigma^2_n(H)$ if it has the form

$$\exists X_1 \forall X_2 \cdots Q X_n \, \varphi,$$

where class quantifiers range over $\mathcal{P}(H)$ and φ is first order over $(H, \in)$, with the class variables as additional predicates. Define $\Pi^2_n(H)$ dually. Unless a domain is displayed, it is HC.

For a real p, a relation is $\Pi^2_n(p)$ if it is defined by a Π^2_n formula with parameter p; define $\Sigma^2_n(p)$ similarly. Without parameters these are the lightface pointclasses. The boldface pointclass $\mathbf{\Pi}^2_n$ is obtained by allowing an arbitrary real parameter. A finite tuple of real parameters is identified with one real by a fixed recursive coding. Parameters displayed as arguments of a formula are always retained as those same arguments in its transformations below.

Definition 2.1. Π^2_n-UP *with the same parameters means that, for every real p and every $\Pi^2_n(p)$ relation $A \subseteq \omega^\omega \times \omega^\omega$, there is a $\Pi^2_n(p)$ relation $U \subseteq A$ with $\operatorname{dom} U = \operatorname{dom} A$ which is the graph of a function. For a lightface A the defining formula of U is also lightface. Define Σ^2_n-UP with the same parameters analogously.*

We prove this parameter-preserving form throughout. It implies both the lightface and boldface uniformization properties. Until Section 10, the extensions constructed for the main theorem add no reals and preserve ω_1. Thus they have the same HC as L, satisfy CH, and have only constructible real parameters. Their collections of classes over HC may differ.

2.2 Cardinals

Until Section 10 we write

$$\kappa_0 = \omega, \qquad \kappa_i = \omega_i^L \ (i \geq 1), \qquad \mu = \aleph_\omega^L, \qquad \lambda = \mu^+, \qquad \Theta = \mu^{++}.$$

The forcings constructed below preserve cardinals, so these are the cardinals ω_i, $\aleph_\omega$, $\aleph_{\omega+1}$, $\aleph_{\omega+2}$ of the final model.

2.3 A criterion for simultaneous uniformization

Fix a parameter-free recursive injective coding $\ell(e, p, x, y) \in 2^\omega$ of a formula index and a real triple, and let I be its range in L. We write $\langle e, p, x, y\rangle$ for this real label. Thus $|I| = \kappa_1$, and distinct sections have disjoint sets of labels. We regard these labels as subsets of ω.

For support sequences write $\vec{R} \leq \vec{T}$ if $R_n \subseteq T_n$ for every n. The following criterion is proved in Section 9.

Theorem 2.2. *Let I be the set of constructible real codes. Suppose that a uniformly definable family of set forcings $F_{\vec{T}} \in L$, indexed by $\vec{T} \in (\mathcal{P}(I))^\omega \cap L$, satisfies:*

(A1) each forcing preserves cardinals, adds no reals, and decides every instance with real parameters of every finite second-order formula;

(A2) for each n there is a parameter-free Σ^2_n formula Code_n such that $F_{\vec{T}} \Vdash (\mathrm{Code}_n(t) \leftrightarrow t \in T_n)$;

(A3) if $\vec{R} \leq \vec{T}$ differ only at n, every Π^2_n instance with real parameters forced by $F_{\vec{T}}$ is forced by $F_{\vec{R}}$;

(A4) if $\vec{R}, \vec{T}$ agree through n, their decided Π^2_n instances with real parameters agree, even if infinitely many higher coordinates differ.

Assume that the relation giving these forcing decisions is uniformly definable in L from the finite formula codes, the sequences of supports, and the real parameters. Then some $\vec{T}_* \in L$ has $F_{\vec{T}_*}$ forcing Π^2_n-UP for every $n \geq 1$, with the same parameters.

2.4 Forcing terminology

We use the forcing and iteration terminology of [9, Chapters 14–16]. A forcing R is $< \rho$-*closed* if every descending sequence of length $< \rho$ has a lower bound, and $< \rho$-*distributive* if it adds no new sequences of ordinals of length $< \rho$. It is $< \rho$-*compact* if every finitely compatible family of fewer than ρ conditions has a lower bound, and ρ-*linked* if it is the union of ρ many sets of pairwise compatible conditions. A forcing is *cone homogeneous* if for any two conditions p, q there are $p' \leq p$, $q' \leq q$ with isomorphic forcings below p' and q', via an isomorphism fixing the relevant ground parameters. A projection $\pi : P \to Q$ preserves the order and the largest condition and, whenever $q \leq \pi(p)$, admits $p' \leq p$ with $\pi(p') \leq q$. For the restriction maps used below we prove the stronger extension property $\pi(p') = q$, with p' agreeing with p outside the retained coordinates. We refer explicitly to this extension property when it is needed. Following [2], a forcing R is *countably parallel-closed* if, whenever $\langle p_n : n < \omega\rangle$ and $\langle q_n : n < \omega\rangle$ are descending sequences in R such that p_n and q_n are compatible for every

$n < \omega$, there is $r \in R$ with $r \leq p_n, q_n$ for all $n < \omega$. For a regular uncountable cardinal ρ, R satisfies the *stationary ρ^+-chain condition* if for every sequence $\langle p_\alpha : \alpha < \rho^+ \rangle$ in R there are a club $C \subseteq \rho^+$ and a regressive function $f : C \cap \{\alpha < \rho^+ : \mathrm{cf}(\alpha) = \rho\} \to \rho^+$ such that p_α and p_β are compatible whenever $\alpha, \beta \in C$ have cofinality ρ and $f(\alpha) = f(\beta)$.

The following form of the preservation theorem follows from [2, Theorem 1.2].

Fact 2.3. *Let ρ be regular uncountable with $\rho^{<\rho} = \rho$. A $< \rho$-support iteration whose iterands are forced to be $< \rho$-compact and to satisfy the stationary ρ^+-chain condition is $< \rho$-closed and satisfies the stationary ρ^+-chain condition, hence the ρ^+-chain condition.*

Compactness gives the closure and countable parallel closure required by [2, Theorem 1.2]. The following implications are recorded in [2, Lemma 1.5]: countable compactness implies countable parallel closure, and ρ-linkedness implies the stationary ρ^+-chain condition. For completeness, two pointwise compatible descending ω-sequences have finitely compatible union, so countable compactness supplies a common bound. Also, on a club of indices above ρ, the value of a map witnessing ρ-linkedness is regressive. A $< \rho$-compact forcing is $< \rho$-closed and countably compact. These checks concern the individual factors; an entire iteration is not asserted to be ρ-linked.

2.5 Club shooting and almost-disjoint coding

For a regular ρ and $S \subseteq \rho$ with $\rho \setminus S$ stationary, $\mathbb{C}(\rho \setminus S)$ is the forcing of closed bounded subsets of $\rho \setminus S$, ordered by end extension.

Fix a regular ρ with $\rho^{<\rho} = \rho$, $\sigma = \rho^+$, and a constructible almost-disjoint family $\langle A_\eta : \eta < \sigma \rangle$ of unbounded subsets of ρ, with pairwise bounded intersections. For $D \subseteq \sigma$, the forcing $\mathbb{A}_\rho(D)$ consists of pairs (s, F) with $s \in 2^{<\rho}$ and $F \in [D]^{<\rho}$, where $(s', F') \leq (s, F)$ iff $s' \supseteq s$, $F' \supseteq F$, and $s'(\alpha) = 0$ whenever $\alpha \in \mathrm{dom}\, s' \setminus \mathrm{dom}\, s$ and $\alpha \in A_\eta$ for some $\eta \in F$. If $Z = \bigcup\{s : (s, F) \in G\}$ for a generic G, then

$$\eta \in D \iff Z \cap A_\eta \text{ is bounded in } \rho.$$

$\mathbb{A}_\rho(D)$ is $< \rho$-compact and ρ-linked (conditions with the same stem are compatible).

2.6 Absoluteness

Lemma 2.4. *Let ρ be regular, R be $< \rho$-distributive and P have a ground-model dense subset of size $< \rho$. Use the fixed ground posets in the product. Then $V[G_P][G_R]$ and $V[G_P]$ have the same sequences of ordinals of length $< \rho$. Equivalently, the old R is $< \rho$-distributive after forcing with P.*

Proof. For $\rho = \omega$ the assertion is immediate. Otherwise, a fixed dense suborder of P of size $< \rho$ remains ρ-cc in V^R. Apply [24, Lemma 4.7]. $\square$

Fact 2.5 (Silver)**.** *Let ρ be regular with $\rho^{<\rho} = \rho$ and let R be $< \rho$-closed. Then $\Sigma^2_1(H_\rho)$ statements with parameters in $\mathcal{P}(H_\rho) \cap V$ are absolute between V and V^R.*

Proof. Since $|H_\rho| = \rho$ and $H_\rho^{V^R} = H_\rho^V$, code $(H_\rho, \in, \vec{A})$ by a structure on ρ in V. Apply Silver's absoluteness lemma for existential second-order formulas [6, Lemma 3.5]; see also [11, p. 298, (I6)]. $\square$

The following relative form of Silver's lemma applies to forcing statements about a small forcing P which may change H_ρ.

Lemma 2.6. *Let ρ be regular uncountable with $\rho^{<\rho} = \rho$, let P be ρ-cc of density at most ρ, and let R be $< \rho$-closed. Fix P-names $\dot{A}_i$ for predicates on $H_\rho^{V^P}$, a first-order formula θ and $p \in P$. Then*

$$V^R \models [p \Vdash_P \exists X \subseteq H_\rho \; \theta(X, \dot{A}_i)] \iff V \models [p \Vdash_P \exists X \subseteq H_\rho \; \theta(X, \dot{A}_i)].$$

Proof. Index a dense suborder $P_0 \subseteq P$, with $p \in P_0$ and $|P_0| \leq \rho$, by ordinals below ρ. Put $H = H_\rho^V$ and $\mathbb{B} = \mathrm{RO}(P)^V$. If $r \Vdash_R \langle \dot{p}_\alpha : \alpha < \rho \rangle$ is an antichain in P, closure gives a descending sequence $\langle r_\alpha : \alpha < \rho \rangle$, $r_0 = r$, with

$$r_{\alpha+1} \Vdash_R \dot{p}_\alpha = \check{p}_\alpha \quad \Longrightarrow \quad p_\alpha \perp_P p_\beta \quad (\alpha < \beta < \rho),$$

contradicting the ρ-cc in V. Thus $R \Vdash P$ is ρ-cc.

Let $\mathcal{N} \subseteq H$ code nice P-names for well-founded extensional relations on $\gamma < \rho$, together with maximal antichains deciding their distinguished points. These codes have hereditary size $< \rho$; a failure of well-foundedness is witnessed by an ω-sequence whose graph has a nice P-name. Since R adds no such codes,

$$H_\rho^{V^R} = H, \qquad \mathcal{N}^{V^R} = \mathcal{N}^V, \qquad |\mathcal{N}| \leq \rho.$$

Coding transitive closures, the maximum principle and mixing show that these names represent every element of the generic H_ρ in both models. Identify each code with the name for its distinguished collapsed element.

Each Boolean value has an antichain code of size $< \rho$. Hence $\mathrm{RO}(P)^{V^R} = \mathbb{B}$ and $|\mathbb{B}| \leq \rho^{<\rho} = \rho$. For $D \subseteq \mathbb{B}$ in V^R, choose a maximal antichain $A \subseteq \{q \in P_0 : \exists d \in D \; (q \leq d)\}$. Then $|A| < \rho$, $A \in V$, and $\bigvee D = \bigvee_{\mathbb{B}} A$; thus $\mathbb{B}$ remains complete.

Index $\mathbb{B}$ below ρ. Fix on H the predicates $\mathcal{N}$, $\mathbb{B}$, its operations, and

$$e(u,v) = \|u = v\|, \qquad m(u,v) = \|u \in v\|, \qquad a_i(u) = \|u \in \dot{A}_i\|.$$

These predicates agree in V and V^R. Predicate names correspond to class tables satisfying

$$\mathrm{Coh}(\chi) \iff \chi : \mathcal{N} \to \mathbb{B} \ \wedge \ \forall u, v \in \mathcal{N}\, [e(u,v) \wedge \chi(u) \leq \chi(v)].$$

For each subformula φ of θ, introduce a valuation table v_φ on finite assignments from $\mathcal{N}$, with

$$\begin{aligned} v_{z=w}(u,v) &= e(u,v), & v_{z\in w}(u,v) &= m(u,v), \\ v_{z\in X}(u) &= \chi(u), & v_{z\in A_i}(u) &= a_i(u), \\ v_{\neg\varphi}(\vec{u}) &= \neg v_\varphi(\vec{u}), & v_{\varphi\wedge\psi}(\vec{u}) &= v_\varphi(\vec{u}) \wedge v_\psi(\vec{u}), \\ v_{\exists z\,\varphi}(\vec{u}) &= \bigvee_{w\in\mathcal{N}} v_\varphi(w,\vec{u}). \end{aligned}$$

The join clause is first order over H: for $b \in \mathbb{B}$,

$$b = \bigvee_w c_w \iff (\forall w\ c_w \leq b) \ \wedge \ \forall d \in \mathbb{B}\big[(\forall w\ c_w \leq d) \to b \leq d\big].$$

Write $\mathrm{Val}_\theta(\chi, \vec{v})$ for these clauses, including the table domains and ranges. The maximum principle gives

$$p \Vdash_P \exists X\, \theta(X, \dot{A}_i) \iff \exists \chi, \vec{v}\, [\mathrm{Coh}(\chi) \wedge \mathrm{Val}_\theta(\chi,\vec{v}) \wedge p \leq v_\theta].$$

Coding the finitely many tables by one subset of H, the right-hand side is $\Sigma^2_1(H)$ with the same ground predicates in V and V^R. Apply Fact 2.5. $\square$

Remark 2.7. The same translation applies to an arbitrary finite prefix of class quantifiers: a universal class block ranges, externally, over all valid name tables. It is essential that higher class quantifiers are not internalised into a transitive model which merely contains H_ρ, as such a model may omit many subsets of H_ρ. For the universal dual, note that $p \Vdash_P \forall X\, \theta(X)$ holds exactly when no $q \leq p$ forces $\exists X\, \neg\theta(X)$. Apply Lemma 2.6 to every such q. This gives the same relative absoluteness for Π^2_1 assertions, retaining the actual condition p.

2.7 Term forcing

Let ρ be regular uncountable, $\sigma = \rho^+$, $\rho^{<\rho} = \rho$ and $\sigma^{<\rho} = \sigma$. Let P be ρ-cc with a fixed enumeration of a dense subset of cardinality at most ρ, and let $P \Vdash \dot{Q} = \mathbb{A}_\rho(\dot{B})$ with $\dot{B} \subseteq \sigma$.

Definition 2.8. *The* full term forcing $\mathrm{Term}(P, \dot{Q})$ *consists of P-names $\tau = (\dot{s}, \dot{F})$ with $1_P \Vdash \tau \in \dot{Q}$, using nice names over the fixed dense suborder for the stem graph and the restraint set. Put $\tau \leq \upsilon$ iff $1_P \Vdash \tau \leq \upsilon$.*

Note that the stems of terms are themselves P-names, not ground stems.

Theorem 2.9. $\mathrm{Term}(P, \dot{Q})$ *is* $< \rho$*-compact,* ρ*-linked and has a dense subset of size at most* σ*. The map* $(p, \tau) \mapsto (p, \tau)$ *is a projection*

$$P \times \mathrm{Term}(P, \dot{Q}) \twoheadrightarrow P * \dot{Q}.$$

Proof. Since P is ρ-cc, every name for an ordinal below ρ has a uniform bound below ρ: use a maximal antichain deciding it and regularity of ρ. A stem name therefore has a length bound below ρ. A nice name for its graph has size $< \rho$, so there are at most $\rho^{<\rho} = \rho$ stem names. Terms with the same nice name for the stem graph are forced compatible by taking the union of their restraint sets. The maximum principle gives a common term extension. Assigning terms their stem-name indices proves ρ-linkedness.

A restraint set has a uniformly bounded size below ρ. Names for an enumeration of it use fewer than ρ indexed entries, with ordinal values below σ. Together with the stem names there are at most $\sigma^{<\rho} = \sigma$ such condition names. Every term has an equivalent name of this kind, giving the required dense subset.

Let $\{\tau_\alpha : \alpha < \eta\}$ be a finitely compatible family of terms, where $\eta < \rho$. For every finite subfamily a term lower bound witnesses that its evaluations are compatible in every P-extension. Hence P forces that the entire displayed family is centered. It also preserves the regular cardinal ρ. Compactness of $\mathbb{A}_\rho(\dot{B})$ gives a name for a lower bound of all the evaluations. By the maximum principle choose this name valid below 1_P, and replace its components by nice names. It is a lower bound in the term forcing, proving $< \rho$-compactness.

Finally suppose $(p', \dot{q}) \leq (p, \tau)$ in $P * \dot{Q}$. Extend $\{p'\}$ to a maximal antichain in P. Mix $\dot{q}$ below p' with τ below the other members of that antichain. The mixed name τ' is forced by 1_P to be below τ, and below p' it equals $\dot{q}$. Thus $(p', \tau') \leq (p, \tau)$ in $P \times \mathrm{Term}(P, \dot{Q})$ and its image is below $(p', \dot{q})$. The order and largest-condition clauses are immediate. This proves that the displayed map is a projection. □

Remark 2.10. The same proof applies to P-names for factors which are $< \rho$-compact and have a specified map witnessing ρ-linkedness. An iteration of such forcings need not be ρ-linked. To apply term forcing to an iteration, lift its iterand names successively along the preceding projections from products to two-step iterations.

Remark 2.11. The term forcing is formed in the model in which $\rho^{<\rho} = \rho$. The later P-generic filter evaluates its terms. All applications below use the conditions and order of the term forcing from that model.

3 Stationary sets and club shooting

For $n \geq 2$ put $\rho = \kappa_{n-1}$ and $\nu = \kappa_n$. The stationary sets at level n will be indexed by the reals in I.

The coding is related to the stationary-set constructions in [7, Section 2] and [4, Section 2, Steps 0–1]. We give the definitions and proofs needed here at every finite level. At level one put $\rho = \omega$ and $\nu = \omega_1$; only the diamond sequence and its stationary sets are needed there.

3.1 Constructible definitions

Fix the standard good Σ_1 Gödel wellorder $<_L$, ordered by construction stage and then by least definition; see [9, Chapter 13]. Write $\mathrm{Init}(p, J)$ for $J = \{q : q <_L p\}$, and fix a bounded formula q_{init} with

$$\mathrm{Init}(p, J) \iff \exists w\, q_{\mathrm{init}}(p, J, w).$$

For $p \in L_\nu$, both J and a witness w belong to L_ν.

For $\delta < \nu$, witness $\mathrm{cf}(\delta) = \rho$ by a strictly increasing cofinal map $\rho \to \delta$, and its negation by $\delta = 0$ or a cofinal map $\gamma \to \delta$ with $\gamma < \rho$. Here cofinal means that every $\eta < \delta$ is at most a value of the map, allowing one-point witnesses for successor ordinals. These witnesses lie in $H^L_\nu = L_\nu$. Collecting them for $\delta \le a$ gives a Σ_1 definition of

$$A_a = (a+1) \cap \mathrm{cof}(\rho) \qquad (a < \nu).$$

We use these witnesses in the bounded formulas below.

3.2 A Σ_1 definition of the diamond sequence

Let $A = \nu \cap \mathrm{cof}(\rho)$. At each $\delta \in A$, choose the $<_L$-least pair (g_δ, c_δ) with $g_\delta \subseteq \delta$, c_δ club in δ, and

$$\forall \eta \in A \cap c_\delta \quad g_\delta \cap \eta \neq g_\eta. \tag{S1}$$

The recursion is defined since $A \cap \delta$ is nonstationary in δ. To verify $\diamondsuit(A)$, suppose that (X, C) is the $<_L$-least counterexample. In a sufficiently large limit L_θ, choose an elementary hull H containing $\rho, \nu, \vec{g}, X, C$, with ρ pointwise included and $\delta = H \cap \nu \in C \cap \mathrm{cof}(\rho)$. The hull has size ρ and, when $\rho > \omega$, may be chosen closed under $< \rho$-sequences. For each $\eta < \delta$, elementarity supplies the relevant cofinality witness in H; since its domain is at most ρ and $\rho \subseteq H$, the collapse fixes that witness. Condensation therefore sends the history to $\vec{g} \restriction \delta$ and the least counterexample to $(X \cap \delta, C \cap \delta)$. The stage-first order contains all predecessors of this pair in the collapsed level, and the local test is absolute. Thus $g_\delta = X \cap \delta$, contradicting $\delta \in C$. This is Jensen's least-counterexample argument [10, Lemma 6.5, pp. 293–294]. The good Σ_1 recursion gives a functional graph Φ_G, with a fixed bounded formula q_G such that

$$\Phi_G(\rho, \delta, g) \iff \exists w\, q_G(\rho, \delta, g, w).$$

A witness records the recursion through δ, its cofinality witnesses and the exact $<_L$-initial segments used to verify leastness; all belong to L_ν.

For $t \subseteq \omega$, put

$$\mathrm{Tr}(g,t) \iff \omega \in g \ \wedge \ g \cap \omega = t,$$

$$S_t = \{\delta \in A : \mathrm{Tr}(g_\delta, t)\}, \qquad S_* = \{\delta \in A : g_\delta = \varnothing\}.$$

Guessing $t \cup \{\omega\}$ proves stationarity of S_t; guessing the empty set proves stationarity of S_*. Distinct S_t are disjoint, and all are disjoint from S_*. The real t is fixed by every collapse fixing ρ.

3.3 A formula asserting the existence of an avoiding club

Define $\psi_{\mathrm{src}}(\rho, \nu, C, t)$ by existentially supplying K, U, F such that:

1. $C \subseteq \nu$ is club in ν;
2. $K \subseteq \nu$ and U assigns, to every $\delta < \nu$, a cofinality witness of the appropriate kind described above: a strictly increasing cofinal map from ρ if $\delta \in K$, and a witness to $\mathrm{cf}(\delta) < \rho$ or to $\delta = 0$ otherwise. Thus $K = \nu \cap \mathrm{cof}(\rho)$;
3. F has domain K, and $F(\delta) = (g_\delta, w_\delta)$ with $q_G(\rho, \delta, g_\delta, w_\delta)$;
4. $\forall \delta \in C \cap K \ \neg \mathrm{Tr}(g_\delta, t)$.

The formula following $\exists K \, \exists U \, \exists F$ is bounded, so ψ_{src} is Σ_1 and asserts that C is club in ν and disjoint from S_t. Its argument is C itself, not the entire set coding C.

The graphs of the diamond sequence and its least witnesses, together with the cofinality witnesses, belong to $L_{\nu+r}$ for some $r < \omega$. The ordinary hierarchy through α and its set-level satisfaction relations can be constructed in $L_{\omega^2 \cdot (\alpha+1)}[C]$: at successors compute satisfaction and Def, and at limits collect the previous hierarchy sequences. Since $\omega^2 \cdot \nu = \nu$, the required witnesses lie in $L_{\nu \cdot 2}[C]$, while $C \in L_{\nu+1}[C]$. Thus, whenever C is an avoiding club,

$$L_{\nu \cdot 2+1}[C] \models \psi_{\mathrm{src}}(\rho, \nu, C, t). \tag{S2}$$

3.4 The constructible almost-disjoint family

Enumerate $\mathcal{P}(\rho)^L$ in $<_L$ as $\langle U_\eta : \eta < \nu \rangle$, identifying subsets with their characteristic functions. Let $h(s)$ be the $<_L$-index of s among constructible binary strings of length $< \rho$; this order has type ρ. Put

$$B_\eta = \{h(U_\eta \restriction \zeta) : \zeta < \rho\}.$$

The good Σ_1 definition of $<_L$, with Collection, gives a functional Σ_1 graph $\Phi_B(\rho, \eta, B)$ over L_ν. Each B_η is unbounded in ρ, and distinct branches give bounded intersections. We always use this fixed family for almost disjoint coding at ρ.

3.5 Coding clubs

For fixed $n \geq 2$, use ZFC without Power Set, retaining Collection and Choice, together with existence of ω_{n+1} and $\omega_{n+1} = \omega_{n+1}^L$. Restrict attention to transitive models N in which the distinguished ρ is the true κ_{n-1}, and put $\beta = (\rho^+)^N$. Require N to satisfy the following additional first-order hypotheses:

1. over its set L_β, Φ_G is total and functional at every $\delta \in \beta \cap \mathrm{cof}(\rho)$;
2. over L_β, Φ_B is total and functional at every $\eta < \beta$;
3. the maps coding the even and odd parts and the fixed pairing functions agree with the displayed canonical definitions.

The transitive model's L_β is the actual L_β. Upward Σ_1 absoluteness and global functionality identify the local definitions of the diamond sequence and the almost-disjoint family with their actual restrictions below β. These additional hypotheses hold in the transitive collapses of the elementary submodels used for the converse implication, by elementarity from a sufficiently large H_Θ.

Use $\epsilon_i(\alpha) = 2 \cdot \alpha + i$ $(i < 2)$, and put

$$\mathrm{Even}(P) = \{\alpha : \epsilon_0(\alpha) \in P\}.$$

Define the odd part using ϵ_1. These maps and their inverses preserve every infinite cardinal, including the internal cardinals of the models under consideration. Hence

$$\mathrm{Even}(P \cap \beta) = \mathrm{Even}(P) \cap \beta,$$

and likewise for the odd part.

For a club $C \subseteq \nu \setminus S_t$, define its padded code as follows. Put $K = L_{\nu \cdot 2+1}[C]$. In $L[C]$, choose canonically a continuous increasing chain of elementary submodels covering K, whose members have cardinality ρ and contain ρ pointwise and ρ, ν, C, t as elements. Let E be the club of their intersections with ν, thinned so that $\min E > \rho$ and successive gaps have length at least ρ. Define $P = P_t^n(C)$ by $\mathrm{Even}(P) = C$ and, for successive $c < d$ in $E \cup \{0\}$, place the $<_L$-least code of a wellorder of ρ of type d in the odd block $[c, c + \rho)$, using the fixed canonical pairing on ρ. All unused odd positions are zero. This fixes the assignment $C \mapsto P_t^n(C)$ uniformly in $L[C]$.

Lemma 3.1. *Let $C \subseteq \nu \setminus S_t$ be club and put $P = P_t^n(C)$. If N is a suitable transitive model satisfying the additional first-order hypotheses above, its distinguished ρ is the true ρ, $\beta = (\rho^+)^N < \nu$, and $P \cap \beta \in N$, then*

$$N \models \psi_{\mathrm{src}}(\rho, \beta, C \cap \beta, t).$$

Proof. If $\beta \notin E$, take successive $c < d$ in $E \cup \{0\}$ with $c < \beta < d$. In N, $c + \rho < \beta$, so $P \cap \beta$ contains the full odd-block code of a wellorder of ρ of type d. Transitivity and the true ρ ensure that N computes its actual order type. Its collapse therefore gives in N a bijection between ρ and $d > \beta$, contradicting $\beta = (\rho^+)^N$. Thus $\beta \in E$. Let H_β be the chosen chain member with $H_\beta \cap \nu = \beta$. Its collapse fixes ρ and t, sends ν to β and C to $C \cap \beta$, and sends the relative hierarchy of K to the hierarchy over $C \cap \beta$. The map $\eta \mapsto \nu + \eta$ identifies the two ordinal blocks in the height of K, so relative condensation gives

$$\bar{K} = L_{\beta \cdot 2+1}[C \cap \beta] \subseteq N.$$

The height is computed using $\eta \mapsto \nu + \eta$ in K; the suitable theory and Collection put this relative level in N. By (S2), elementarity and upward Σ_1 absoluteness,

$$\bar{K} \models \psi_{\mathrm{src}}(\rho, \beta, C \cap \beta, t) \quad \Longrightarrow \quad N \models \psi_{\mathrm{src}}(\rho, \beta, C \cap \beta, t).$$

□

If $Z \subseteq \rho$ almost-disjoint codes P, then in each such N containing Z the decoded set is $P \cap \beta$: Φ_B gives the actual family and boundedness in the true ρ is absolute. Separation puts $P \cap \beta$ in N, so Lemma 3.1 applies.

Conversely, test a proposed code against collapses of elementary submodels of a sufficiently large H_Θ, of size ρ, containing ρ pointwise, Z, and the actual constructible definitions and their satisfaction relations. Their intersections with ν are cofinal in ν. Condensation identifies their L_β with the actual L_β, and the collapse fixes Z and every B_η for $\eta < \beta$. If all these models satisfy the local conclusion, the even part of the decoded set has club restrictions avoiding S_t at cofinally many β; it is therefore itself an avoiding club.

3.6 The product of the club forcings

For a ground-model sequence $\vec{T} = \langle T_n : n \geq 1 \rangle$ with $T_n \subseteq I$, let $C_n(T_n)$ be the product of the forcings $\mathbb{C}(\kappa_n \setminus S_t^n)$, $t \in T_n$, with countable support for $n = 1$ and full support (over the κ_1 label coordinates) for $n \geq 2$. Let

$$C_{\vec{T}} = \prod_{n \geq 1} C_n(T_n) \quad \text{(full support)}, \qquad B_{\vec{T}} = L[G_{C_{\vec{T}}}].$$

All conditions are ground conditions; in particular higher factors are not recomputed after lower generic objects appear. For $\vec{R} \leq \vec{T}$ (coordinatewise inclusion), $C_{\vec{R}}$ is a complete subforcing of $C_{\vec{T}}$ via the natural product projection.

Lemma 3.2. *For every $n \geq 1$ and $T \subseteq I$, the forcing $C_n(T)$ has cardinality at most κ_n, is $< \kappa_n$-distributive, and preserves the stationarity of S^n_* and of every S^n_t with $t \notin T$. For $n \geq 2$ it is also $< \kappa_{n-1}$-closed.*

Proof. Put $\rho = \kappa_{n-1}$ and $\nu = \kappa_n = \rho^+$. The support has size at most ρ: it is countable for $n = 1$, and contained in I, $|I| = \kappa_1 \leq \rho$, otherwise. GCH gives

$$|C_n(T)| \leq (\nu^\rho)^{|I|} = \nu \quad (n \geq 2), \qquad |C_1(T)| \leq \omega_1^\omega = \omega_1.$$

For $n \geq 2$, a descending sequence of length $< \rho$ has a lower bound coordinatewise: take the unions and append their suprema. A new supremum has cofinality $< \rho$ and so belongs to no $S^n_t \subseteq \mathrm{cof}(\rho)$.

Let S be either the reserve or an omitted stationary set. Work below $p \in C_n(T)$. Given the required dense sets, choose

$$M \prec H_\theta, \quad |M| = \rho, \quad \rho \subseteq M, \quad p, C_n(T), \vec{S} \in M, \quad \delta = M \cap \nu \in S.$$

For $\rho > \omega$ require $M^{<\rho} \subseteq M$. Such models are obtained from a continuous elementary chain of length ν: at indices of cofinality ρ take unions of $< \rho$-closed hulls; the intersections with ν form a club. For $\rho = \omega$ use countable hulls. Externally enumerate the dense open subsets of $C_n(T) \restriction p$ belonging to M in length ρ, and fix $\langle \delta_i : i < \rho \rangle$ cofinal in δ, with $\delta_i \in M \cap \nu$ for every i.

Build a descending sequence $\langle p_i : i < \rho \rangle$ in M starting below p and meeting these dense sets. After each dense set, synchronize the last points of all active coordinates at a common $\gamma_i \in S^n_* \cap M$, above all their previous last points and the next chosen ordinal below δ. The bound on the previous last points is definable from the single condition $p_i \in M$, hence is in $M \cap \nu$. Elementarity and unboundedness of the reserve supply γ_i. At limits $i < \rho$, closure and $M^{<\rho} \subseteq M$ give a lower bound still in M. In the countable-support case, also activate the coordinates in $T \cap M$ successively. The union of the supports has the required size. Put

$$q(t) = \bigcup_{i<\rho} p_i(t) \cup \{\delta\} \qquad (t \text{ in this union of supports}).$$

Every active coordinate has last points cofinal in δ. Since $\delta \in S$ and S is disjoint from all selected S^n_t, q is a condition below every p_i.

To prove $< \nu$-distributivity, include any given sequence of fewer than ν dense open sets and its length η in M. Since $|\eta| \leq \rho$ and $\rho \subseteq M$, elementarity gives $\eta \subseteq M$. Its entries all belong to M, and the constructed q meets each

of them. To preserve S, include a name $\dot{C}$ for a club in ν in M. For each $\alpha \in M \cap \nu$, the dense set deciding a point of $\dot{C}$ above α belongs to M. Its meeting condition has a witness in $M \cap \nu$ by elementarity. Hence q forces $\dot{C} \cap \delta$ cofinal in δ, and therefore $\delta \in \dot{C} \cap S$. □

Lemma 3.3. *$C_{\vec{T}}$ preserves all cardinals and the GCH and adds no countable sequences of ordinals. Every S^n_t with $t \notin T_n$ remains stationary. $C_{\vec{T}}$ is cone homogeneous.*

Proof. Lemma 3.2 gives $|C_n(T_n)| \leq \kappa_n$, $< \kappa_n$-distributivity and preservation of the reserve and all omitted stationary sets. For $n \geq 2$ it also gives $< \kappa_{n-1}$-closure.

Fix n. The product above n is $< \kappa_n$-closed, preserving H_{κ_n} and those stationary sets. The old $C_n(T_n)$ therefore still has all its conditions, and the same master-condition argument applies. The product below n has size $< \kappa_n$, so it preserves the stationarity of each omitted S^n_t. Lemma 3.4 gives cardinal preservation, GCH and the absence of new countable sequences.

For cone homogeneity, extend two conditions to equal active supports and, at each level, a common endpoint in S^n_*. Replacing one initial segment by the other and keeping the tail defines a cone isomorphism. Their product fixes all ground-model parameters. □

Lemma 3.4. *Assume the single-factor distributivity and stationarity properties used in the proof of Lemma 3.3. Then its full product $C_{\vec{T}}$ preserves all cardinals and GCH, and adds no countable sequences of ordinals.*

Proof. Write $Q_{\geq n} = \prod_{m \geq n} C_m(T_m)$ and $Q_{<n} = \prod_{m<n} C_m(T_m)$. The $< \kappa_n$-closure of $Q_{>n}$ preserves H_{κ_n} and the reserve, so the same fusion argument for $C_n(T_n)$ gives

$$Q_{\geq n} \text{ is } < \kappa_n\text{-distributive}, \qquad |Q_{<n}| < \kappa_n.$$

Hence every κ_n remains regular, and $n = 1$ gives no new countable ordinal sequences. Also, in L,

$$|C_{\vec{T}}| \leq \mu^\omega \leq 2^\mu = \lambda,$$

so $C_{\vec{T}}$ is Θ-cc. The finite factorizations give $\mathrm{cf}(\lambda) \geq \kappa_n$ for every n, by the antichain argument in Lemma 4.8. Since $\mu = \sup_n \kappa_n$ remains singular, λ remains a regular cardinal. Together with the Θ-cc this proves preservation of all cardinals.

For $i \geq 1$, $Q_{>i}$ adds no subsets of κ_i and $|Q_{\leq i}| \leq \kappa_i$. Counting names therefore gives

$$2^{\kappa_i} \leq \left((2^{\kappa_i})^{\kappa_i}\right)^L = \kappa_{i+1}.$$

CH follows from the absence of new reals. In the extension fix bijections $e_i : \kappa_{i+1} \to \mathcal{P}(\kappa_i)$ $(i \geq 1)$. The assignment

$$X \longmapsto \langle e_i^{-1}(X \cap \kappa_i) : i \geq 1 \rangle \qquad (X \subseteq \mu)$$

is injective and takes values in $(\prod_{i\geq 1}\kappa_{i+1})^L$, since no countable ordinal sequences were added. Thus $2^\mu \leq (\mu^\omega)^L \leq \lambda$, hence $2^\mu = \lambda$. Finally, for every ground-model cardinal $\tau \geq \lambda$, names coded by subsets of $\tau \times C_{\vec{T}}$ give $2^\tau \leq (2^{\tau\cdot\lambda})^L = (\tau^+)^L$. □

Proposition 3.5. *In L, for every $n \geq 1$ there are, uniformly definably in n:*

(S1) pairwise disjoint stationary sets $S^n_t \subseteq \kappa_n \cap \mathrm{cof}(\kappa_{n-1})$, $t \in I$, and a disjoint stationary reserve S^n_*, obtained as the real traces of the canonical constructible diamond sequence on $\kappa_n \cap \mathrm{cof}(\kappa_{n-1})$;

(S2) for $n = 1$, the graph $t \mapsto S^1_t \cap \alpha$ ($\alpha < \omega_1$) is first-order definable over HC *of any outer model with the same reals, using countable constructible construction sequences;*

(S3) for $n \geq 2$, a definable assignment $C \mapsto P^n_t(C) \subseteq \kappa_n$, computed in $L[C]$ from a club C disjoint from S^n_t, which codes C together with the additional initial-segment data of Lemma 3.1. If $\rho = \kappa_{n-1}$ and a suitable transitive model N satisfying the additional hypotheses in Section 3.5 has true ρ, internal successor $\beta = (\rho^+)^N < \kappa_n$, and $P^n_t(C) \cap \beta \in N$, then

$$N \models \psi_{\mathrm{src}}\big(\rho, \beta, \mathrm{Even}(P^n_t(C) \cap \beta), t\big);$$

(S4) in the extension by the club product of Section 3.6, every S^n_t with $t \notin T_n$ remains stationary, and this is witnessed by a master-condition argument using the reserve.

These properties follow from the definitions above, Lemma 3.1, and Lemmas 3.2 and 3.3.

3.7 The case $n = 2$

Before defining the coding forcing, we explain how its later properties give uniformization at level two. Put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$, where $D_{\vec{T}}$ is the coding forcing constructed in the next two sections. Fix a Π^2_2 formula

$$A(p,x,y) \quad\Longleftrightarrow\quad \forall X\, \exists Y\, \theta(p,x,y,X,Y),$$

where θ is first order over HC. Proposition 6.7 gives a $\Pi^2_1(H_{\kappa_2})$ formula Φ_A, independent of $\vec{T}$, such that

$$B_{\vec{T}}[G_{D_{\vec{T}}}] \models A(p,x,y) \quad\Longleftrightarrow\quad B_{\vec{T}} \models \Phi_A(p,x,y).$$

This reduction uses the bounds on names and the bookkeeping properties of $D_{\vec{T}}$ to account for the classes added by the coding forcing.

Decreasing only T_2 leaves H_{κ_2} of the club-forcing extension unchanged. Downward absoluteness of the universal second-order formula Φ_A then preserves a chosen instance of A. Changing supports above level two preserves its truth in both directions. These are the special cases of Propositions 8.2 and 8.3 that will be used in choosing the uniformizer.

The coding predicate has the form

$$\mathrm{Code}_2(t) \quad \Longleftrightarrow \quad \exists X\, \forall Y\, [V_1(Y; X, t) \longrightarrow D_1(Y, t)],$$

where V_1 and D_1 are first order over HC. The predicate V_1 checks a direct code for a suitable transitive model containing X and t; D_1 reads its assertion that the almost disjoint decoding of X gives an avoiding club up to the model's successor of ω_1. Thus Code_2 is Σ^2_2. If $t \in T_2$, the added club is padded and coded into such an X. If $t \notin T_2$, a proposed witness to $\mathrm{Code}_2(t)$, tested against cofinally many model codes, would give a club disjoint from the stationary set S^2_t. Proposition 7.8 proves that

$$F_{\vec{T}} \Vdash (\mathrm{Code}_2(t) \longleftrightarrow t \in T_2).$$

Now list the formulas $A_{2,e}(p, x, y)$ and begin with $T_2 = I$, keeping all other supports fixed. At each step choose the $<_L$-least unassigned section (e, p, x) which the current forcing makes nonempty, and its $<_L$-least possible value y. Remove only $\ell(e, p, x, y)$ from T_2. Every value already chosen remains in its section when the support is decreased, including at intersections of supports. Since there are only κ_1 sections, the recursion terminates before κ_2. At termination every nonempty section has exactly one removed label. Consequently

$$U_{2,e}(p, x, y) \quad \Longleftrightarrow \quad A_{2,e}(p, x, y) \wedge \neg\, \mathrm{Code}_2(\ell(e, p, x, y))$$

is a $\Pi^2_2(p)$ uniformization. No label has been removed at an unassigned section, so the formula is undefined there. Later changes above level two preserve the decided instances of both $A_{2,e}$ and Code_2, and hence preserve this graph and its domain. Section 9 carries out the same recursion successively for every finite n.

4 The coding forcing and its quotients

Fix $\vec{T}$ and write $B = B_{\vec{T}}$. In this section the ground model is B; a ground-model name or code need not belong to L.

4.1 Small iterations

Definition 4.1. *For $i \geq 1$ put $\sigma_i = \kappa_{i+1}$. A* code for a small Γ_i-iteration *specifies a well-order $(J, \prec)$, $J \subseteq \sigma_i$, and a $< \kappa_i$-support iteration*

$$\langle P_\alpha, \dot{Q}_\alpha : \alpha < \delta \rangle, \qquad \delta = \mathrm{otp}(J, \prec),$$

such that, for $\alpha < \delta$,

$$P_\alpha \Vdash \text{“}\dot{Q}_\alpha = (\check{\sigma}_i, \dot{\leq}_\alpha) \text{ is a } < \kappa_i\text{-compact, } \kappa_i\text{-linked preorder.”}$$

Distinct indices may represent equivalent conditions; unused indices represent the largest condition. The code also fixes

$$e_\alpha : \sigma_i \twoheadrightarrow D_\alpha \subseteq P_\alpha, \qquad D_\alpha \text{ dense in } P_\alpha \quad (\alpha \leq \delta),$$

and codes for nice names using these enumerations. These data and each chosen predicate name admit codes c *with* $|c| \leq \sigma_i$. *We include* $P_\delta \restriction p$ *for* $p \in P_\delta$, *and forcings with specified dense embeddings witnessing equivalence to such iterations. The full quotient forcing at level* i *can have more stages than a small iteration.*

Countably compact factors are countably parallel-closed. After replacing each iterated forcing by its constituent factors, Fact 2.3 applies. Thus the small Γ_i-iterations satisfy the κ_{i+1}-chain condition.

4.2 Coordinates, dependence of names, and conditions

Definition 4.2. *The coding forcing* $D = D_{\vec{T}}$ *is defined by a recursion of length* Θ. *Each coordinate* $\xi < \Theta$ *is assigned a* level $i(\xi) \geq 1$, *a ground-model set* $m(\xi) \subseteq \xi$, *and a* $D \restriction m(\xi)$*-name* $\dot{Q}_\xi$ *for a* $< \kappa_{i(\xi)}$*-compact,* $\kappa_{i(\xi)}$*-linked preorder on* $\sigma_{i(\xi)}$, *with the specified bound functions and a map witnessing linkedness. A small iteration selected by the bookkeeping function is replaced by its constituent iterands at separate coordinates. Put*

$$I_k = \{\xi < \Theta : i(\xi) = k\}.$$

The sets on which these names depend satisfy

$$m(\xi) \subseteq \bigcup_{k \geq i(\xi)} I_k,$$
$$\eta \in m(\xi) \Longrightarrow m(\eta) \subseteq m(\xi),$$
$$|m(\xi) \cap I_k| \leq \kappa_{k+1} \quad (k \geq i(\xi)).$$

A set $C \subseteq \Theta$ *is* closed under the dependence of the names *if* $\xi \in C$ *implies* $m(\xi) \subseteq C$. *We subsequently refer to such a* C *as a closed set of coordinates.*

The set $m(\xi)$ may have cardinality μ. The name $\dot{Q}_\xi$ depends only on coordinates of the same or a higher level. Codes for names and partial orders involving later coordinates of lower level may nevertheless occur as ground-model parameters.

Definition 4.3. *Fix $\xi \in I_i$, with 0 indexing the largest condition of $\dot{Q}_\xi$. For $\eta < \kappa_i$, let $b_{\xi,\eta}$ be the $D \restriction m(\xi)$-name for the operation on $\vec{a} = \langle a_\zeta : \zeta < \eta \rangle \in \sigma_i^\eta$ given by*

$$\vec{a} \longmapsto \begin{cases} \min\{b < \sigma_i : \forall \zeta < \eta \ (b \leq_{\dot{Q}_\xi} a_\zeta)\}, & \vec{a} \text{ descending}, \\ 0, & \text{otherwise}. \end{cases}$$

Let $\mathcal{N}_\xi$ consist of the names represented by well-founded term trees t with $|t| < \kappa_i$, using only the constants $\check{a}$ $(a < \sigma_i)$ and operations $b_{\xi,\eta}$ $(\eta < \kappa_i)$. Then $|\mathcal{N}_\xi| \leq \sigma_i$ and $1_{D \restriction m(\xi)} \Vdash \tau \in \dot{Q}_\xi$ for every $\tau \in \mathcal{N}_\xi$.

The conditions of D are the functions p satisfying

$$\operatorname{supp}(p) \subseteq \Theta, \qquad p(\xi) \in \mathcal{N}_\xi \quad (\xi \in \operatorname{supp}(p)),$$
$$|\operatorname{supp}(p) \cap I_k| < \kappa_k \quad (k \geq 1).$$

With $p(\xi) = \check{0}$ for $\xi \notin \operatorname{supp}(p)$, put

$$q \leq_D p \iff \forall \xi \in \operatorname{supp}(p) \cup \operatorname{supp}(q) \quad q \restriction m(\xi) \Vdash_{D \restriction m(\xi)} q(\xi) \leq_{\dot{Q}_\xi} p(\xi).$$

The support of a condition has cardinality at most μ and may meet infinitely many levels. For $j \geq 1$, let $D_{\geq j}$ be the restriction to $\bigcup_{k \geq j} I_k$, and put $D_{>j} = D_{\geq j+1}$. For a closed set of coordinates C, let $D_C = D \restriction C$. We prove below that these restrictions are complete subforcings.

We initially use preorders and then take their separative quotients.

The next lemmas apply to every sequence of iterands satisfying the stated hypotheses, and hence to every initial segment of the construction.

4.3 Complete subforcings and closure

Lemma 4.4. *For every initial segment of the construction and every closed set A of its coordinates, restriction to a closed $B \subseteq A$ is a projection onto D_B. More precisely, if $q \leq p \restriction B$, there is $p' \leq p$ with $p' \restriction B = q$ and $p' \restriction (A \setminus B) = p \restriction (A \setminus B)$. If all coordinates in A have level at least j, every descending sequence in D_A of length $< \kappa_j$ has a lower bound whose restriction to B is any prescribed lower bound of the restricted sequence.*

Proof. Induct on the length of the construction, simultaneously for all closed sets of coordinates. Given $q \leq p \restriction B$, let p' equal q on B and p elsewhere. At a coordinate in B, the entire set on which its name depends lies in B; outside B its component name is unchanged, so the required comparison is reflexive. Thus $p' \leq p$ and $p' \restriction B = q$. Restriction preserves order by the same induction. Transitivity follows from monotonicity of forcing over each $D_{m(\xi)}$ and transitivity of the order of the iterand.

Let $\langle p_\zeta : \zeta < \eta \rangle$ be descending, where $\eta < \kappa_j$, and let r be a lower bound of its restrictions to B. Set $q \restriction B = r$. For $\xi \in A \setminus B$ put

$$q(\xi) = b_{\xi,\eta}(\langle p_\zeta(\xi) : \zeta < \eta \rangle).$$

Regularity keeps both the support at each level and the number of symbols in each component name below the corresponding cardinal. Induction on ξ shows that $q \restriction m(\xi)$ is below all $p_\zeta \restriction m(\xi)$. Hence it forces the displayed input sequence to be descending, so $b_{\xi,\eta}$ returns a lower bound below that condition. Therefore $q \leq p_\zeta$ for all ζ, with $q \restriction B = r$. No chain condition is used. □

If τ is regular and R is $< \tau$-closed, then G_R is $< \tau$-directed in $V[G_R]$. Indeed, for $X \in [G_R]^{<\tau}$, we have $X \in V$, since R adds no ordinal sequences of length $< \tau$. Choose $r \in G_R$ with $r \Vdash \check{X} \subseteq \dot{G}_R$. For each $a \in X$, the set $\{q \leq r : q \leq a\}$ is dense below r. By $< \tau$-closure, their intersection is dense below r and hence meets G_R.

Consequently, if $B \subseteq A$ are closed sets of coordinates of levels at least j, then D_A/G_B is $< \kappa_j$-closed in $V[G_B]$. To see this, let $\langle p_\zeta : \zeta < \eta \rangle$ be descending in D_A/G_B, where $\eta < \kappa_j$. The sequence belongs to V, and there is $r \in G_B$ with $r \leq p_\zeta \restriction B$ for all $\zeta < \eta$. Applying Lemma 4.4 with restriction to B prescribed to equal r gives a lower bound in D_A/G_B.

4.4 The underlying set and order of the quotient

Fix i, put $\kappa = \kappa_i$, $\sigma = \kappa_{i+1}$ and $R = D_{>i}$. By Lemma 4.4, R is $< \sigma$-closed. Work in $W = V[G_R]$. For an initial segment γ of the level-i coordinates, let S_γ be the set in V of the specified level-i tuples on that segment. Define

$$q \preceq_{G_R} p \quad \iff \quad \exists r \in G_R \; (r \cup q \leq_D p).$$

A tuple supported at level i has upper component 1_R. The extension property of the restriction projection identifies this preorder, up to forcing equivalence, with the quotient obtained from all coordinates of level greater than i together with the chosen initial segment at level i. Every quotient condition is equivalent to its restriction to level i, after strengthening its upper component within G_R.

If $C \subseteq I_i$ and $U_i \cup C$ is closed, where $U_i = \bigcup_{k>i} I_k$, these coordinates yield a complete subforcing of the quotient: apply the restriction projection to $U_i \cup C$ and then factor the common forcing R. The underlying set of tuples is unchanged. Each tuple, together with the trees defining its component names, has $< \kappa$ entries. A new such tuple in W would therefore give a new ordinal sequence of length $< \sigma$.

For a level-i coordinate ξ, let A consist of all coordinates of level greater than i and the level-i coordinates preceding ξ. Then A is closed and contains $m(\xi)$. By the closure argument above, the extension from $V[G_{m(\xi)}]$ to

$V[G_A]$ is by a $< \kappa$-closed quotient. Every $< \kappa$-sized family of conditions of the iterand, identified by their ordinal indices, already belongs to $V[G_{m(\xi)}]$. Its compactness is preserved, as are the linking map and witnesses to compatibility. Consequently an iterand which is $< \kappa$-compact and κ-linked in $V[G_{m(\xi)}]$ has these properties over the full preceding level-i forcing. We use the same order and bound functions from the smaller extension.

Proposition 4.5. *For every initial segment γ, the quotient $(S_\gamma, \preceq_{G_R})$ is forcing equivalent in W to the ordinary $< \kappa$-support iteration P_γ of the same iterands, and has the stationary σ-chain condition.*

Proof. Induct on γ, constructing P_γ and interpreting S_γ coordinatewise in it. At a successor coordinate ξ, completeness of $D_{m(\xi)}$ and the induction hypothesis transfer $\dot{Q}_\xi$ to the preceding ordinary iteration. The closure argument above preserves its compactness and linkedness.

Interpretation preserves $\preceq_{G_R}$. Conversely, let $p, q \in S_\gamma$ with $p \leq_{P_\gamma} q$, and put $A = \mathrm{supp}(p) \cup \mathrm{supp}(q)$ and $C_\xi = m(\xi) \cap I_i$. The restriction $p \restriction C_\xi$ already forces the component comparison: an extension forcing its negation could otherwise be amalgamated with p. By induction and the forcing theorem for R, choose $r_\xi \in G_R$ such that

$$(r_\xi \cup (p \restriction C_\xi)) \restriction m(\xi) \Vdash_{D_{m(\xi)}} p(\xi) \leq_{\dot{Q}_\xi} q(\xi) \quad (\xi \in A).$$

Since $|A| < \kappa$ and G_R is $< \sigma$-directed, choose $r \in G_R$ below all r_ξ. Then $r \cup p \leq_D q$, so $p \preceq_{G_R} q$.

For density at a successor ξ, strengthen the preceding part of p to decide $p(\xi) = \check{c}$, $c < \sigma$. Inductive density gives a tuple q with $q \cup \{(\xi, \check{c})\} \leq p$; $1 \Vdash \check{c} \in \dot{Q}_\xi$. At a limit γ with $\mathrm{cf}(\gamma) \geq \kappa$, we have $\sup \mathrm{supp}(p) < \gamma$, so induction applies.

If $\tau = \mathrm{cf}(\gamma) < \kappa$, choose a continuous increasing $\langle \gamma_\eta : \eta \leq \tau \rangle$ with $\gamma_0 = 0$ and $\gamma_\tau = \gamma$, and construct

$$q_\eta \in S_{\gamma_\eta}, \qquad q_\eta \leq_{P_{\gamma_\eta}} p \restriction \gamma_\eta \quad (\eta \leq \tau),$$
$$q_\zeta \preceq_{G_R} q_\eta \quad (\eta < \zeta \leq \tau),$$

with omitted coordinates assigned the largest condition. Start with $q_0 = 1$; at successors apply density below $q_\eta \cup (p \restriction [\gamma_\eta, \gamma_{\eta+1}))$. At a limit $\nu \leq \tau$, take a common lower bound $r \in G_R$ of witnesses for the $< \kappa$ comparisons $q_\zeta \preceq_{G_R} q_\eta$ $(\eta < \zeta < \nu)$. Then $\langle r \cup q_\eta : \eta < \nu \rangle$ is descending in D and belongs to V, since R adds no $< \sigma$-sequences of ground-model objects. Lemma 4.4 yields

$$s \restriction U_i = r, \qquad s \leq_D r \cup q_\eta \quad (\eta < \nu).$$

Put $q_\nu = s \restriction I_i \in S_{\gamma_\nu}$. Continuity gives $q_\nu \leq_{P_{\gamma_\nu}} p \restriction \gamma_\nu$; in particular, $q_\tau \leq_{P_\gamma} p$.

Finally, R preserves $\kappa^{<\kappa} = \kappa$ and $\sigma = \kappa^+$. Compactness and linkedness give $< \kappa$-closure, countable parallel-closure and the stationary σ-chain condition for every iterand. Apply Fact 2.3 to P_γ. $\square$

A coded small iteration may be treated as one iterand if its enumerated dense subset and bound functions are included. Its individual iterands are preserved successively through the same closed quotients. To apply the chain-condition theorem, replace each such iterand by its defining iteration and concatenate these iterations, extending internal lengths if necessary by trivial forcings. Regularity of κ_i keeps each support below κ_i. We do not require the whole small iteration to be κ_i-linked or compact.

Remark 4.6. A closed set of coordinates of D can include higher-level coordinates on which its iterand names depend. After forcing with $D_{>i}$, its level-i part determines a complete subforcing of $D_{\geq i}/G_{D_{>i}}$. More explicitly, for $C \subseteq I_i$ with $U_i \cup C$ closed, where $U_i = \bigcup_{k>i} I_k$, this subforcing is

$$(D \restriction (U_i \cup C))/G_{D_{>i}}.$$

The set C alone need not be closed in D. All assertions below about the size of these lower forcings and the genericity of their filters are made in the appropriate extension by the upper forcing.

4.5 Size and preservation

Lemma 4.7. *Assume $B \models \mathrm{GCH}$, and put $\mu = \aleph_\omega$, $\lambda = \mu^+$ and $\Theta = \lambda^+$. Let D use the component names and supports of Section 4, with any bookkeeping function. Then*

$$|D| \leq \Theta, \qquad D \text{ is } \Theta\text{-cc.}$$

Proof. GCH gives

$$\mu^\mu = 2^\mu = \lambda, \qquad \lambda^\mu = \lambda, \qquad \Theta^\mu = \Theta,$$
$$|\mathcal{N}_\xi| \leq \kappa_{i+1}^{<\kappa_i} = \kappa_{i+1} < \mu \quad (\xi \in I_i).$$

Since $|\operatorname{supp}(p)| \leq \mu$, we obtain $|D| \leq \Theta^\mu \cdot \mu^\mu = \Theta$.

Given $\langle p_\alpha : \alpha < \Theta \rangle$ in D, put

$$A_\alpha = \operatorname{supp}(p_\alpha), \qquad U_\alpha = \bigcup_{\beta<\alpha} A_\beta, \qquad S = \{\alpha < \Theta : \operatorname{cf}(\alpha) = \lambda\}.$$

For $\alpha \in S$, choose $f(\alpha) < \alpha$ with $A_\alpha \cap U_\alpha \subseteq U_{f(\alpha)}$; this is possible since $|A_\alpha \cap U_\alpha| \leq \mu < \operatorname{cf}(\alpha)$. Fodor's lemma gives a stationary $S' \subseteq S$ with $f[S'] = \{\gamma\}$. As $|U_\gamma| \leq \lambda$ and $|[U_\gamma]^{\leq\mu}| \leq \lambda^\mu = \lambda < \Theta$, there are $H \in [S']^\Theta$ and a such that $A_\alpha \cap U_\alpha = a$ for all $\alpha \in H$. Hence

$$a \subseteq A_\alpha \cap A_\beta \subseteq A_\beta \cap U_\beta = a \quad (\alpha < \beta,\ \alpha, \beta \in H).$$

There are at most $\mu^{|a|} \le \lambda$ restrictions $p_\alpha \restriction a$. Thin to $K \in [H]^\Theta$ with $p_\alpha \restriction a = u$ for all $\alpha \in K$. For distinct $\alpha, \beta \in K$, the union $r = p_\alpha \cup p_\beta$ is a function and

$$|\operatorname{supp}(r) \cap I_i| \le |A_\alpha \cap I_i| + |A_\beta \cap I_i| < \kappa_i \quad (i \ge 1).$$

Since $r(\xi) = p_\epsilon(\xi)$ whenever $\epsilon \in \{\alpha, \beta\}$ and $\xi \in A_\epsilon$, every comparison is reflexive and forced by 1; hence $r \in D$ and $r \le_D p_\alpha, p_\beta$. □

Lemma 4.8. *Suppose B has the cardinals of Section 2 and D is Θ-cc. For every $j \ge 1$, assume a factorization $D_{\ge j} * \dot{P}_{<j}$ of D with*

$$D_{\ge j} \text{ is } < \kappa_j\text{-closed}, \qquad D_{\ge j} \Vdash P_{<j} \text{ is } \kappa_j\text{-cc}.$$

Then D preserves all cardinals and adds no reals. The forcing of Section 4 satisfies these hypotheses.

Proof. Let $G \subseteq D$ be generic. Closure and the chain condition preserve each κ_j and its regularity. Thus, in $B[G]$,

$$\operatorname{cf}(\kappa_j) = \kappa_j \quad (j \ge 1),$$
$$\mu = \sup_{j \ge 1} \kappa_j \text{ is a cardinal}, \qquad \operatorname{cf}(\mu) = \omega.$$

Fix j and work in $M = B[G_{\ge j}]$. Closure gives $\operatorname{cf}^M(\lambda) \ge \kappa_j$. If $p \Vdash_{P_{<j}} \dot{f} : \eta \to \lambda$, $\eta < \kappa_j$, choose maximal deciding antichains below p with value sets X_ζ ($\zeta < \eta$). In M,

$$|X_\zeta| < \kappa_j, \qquad X = \bigcup_{\zeta < \eta} X_\zeta \in [\lambda]^{<\kappa_j}, \qquad \sup X < \lambda,$$

and $p \Vdash \operatorname{ran}(\dot{f}) \subseteq X$. Consequently, in $B[G]$,

$$\operatorname{cf}(\lambda) \ge \kappa_j \quad (j \ge 1), \qquad \operatorname{cf}(\lambda) > \mu,$$

since μ is singular. Forcing creates no cardinals between the ground-model cardinals μ and λ, so $\operatorname{cf}(\lambda) = \lambda$. The Θ-cc preserves all cardinals $\ge \Theta$. Also $P_{<1} = 1$ and $D = D_{\ge 1}$ is countably closed, so $\mathbb{R}^{B[G]} = \mathbb{R}^B$.

For the forcing of Section 4, closure and the Θ-cc were proved above. Proposition 4.5 factors $P_{<j}$ into successive quotients at levels $j-1, j-2, \dots, 1$, each κ_{k+1}-cc and hence κ_j-cc ($k < j$). Each preserves κ_j, so their finite iteration is κ_j-cc. □

Lemma 4.9. *Let $\kappa > \omega$ be regular, $\sigma = \kappa^+$ and $\kappa^{<\kappa} = \kappa$. Suppose R is $< \kappa$-closed and preserves σ. Let $Q = \langle Q_\alpha, \dot{S}_\alpha : \alpha < \delta \rangle$ be a $< \kappa$-support iteration with specified linking maps and lower-bound operations, and*

$$Q_\alpha \Vdash \text{“}\dot{S}_\alpha = (\check{\sigma}, \dot{\le}_\alpha) \text{ is } < \kappa\text{-compact and } \kappa\text{-linked.”}$$

In $W = V[G_R]$, the old order Q^V is forcing equivalent to the iteration $\widehat{Q}$ of the same names and orders; its iterands retain these properties. If $|\delta| \le \sigma$ and each specified dense subset has size $\le \sigma$, then Q^V and its specified principal ideals remain in the small forcing class.

Proof. Since R and every Q_α add no $< \kappa$-sequences of ordinals,

$$R \Vdash Q_\alpha^V \text{ is } {<}\kappa\text{-closed}, \qquad Q_\alpha \Vdash R^V \text{ is } {<}\kappa\text{-closed}.$$

Inductively construct the coordinatewise maps $e_\alpha : Q_\alpha^V \to \widehat{Q}_\alpha$. Atomic forcing for fixed old names and orders is absolute, so these maps preserve and reflect order. In the product $R^V \times Q_\alpha^V$, closure of R^V over $V[G_{Q_\alpha}]$ gives

$$[\sigma]^{<\kappa} \cap V[G_{Q_\alpha}][G_R] = [\sigma]^{<\kappa} \cap V[G_{Q_\alpha}].$$

The iterand order and linking map on σ are unchanged, and finite compatibility is absolute, with ordinal indices as witnesses. Thus compactness and linkedness are preserved.

For $p \in \widehat{Q}_{\alpha+1}$, choose $p' \le p \restriction \alpha$ deciding $p(\alpha) = \check{c}$, $c < \sigma$, and $q \in Q_\alpha^V$ with $e_\alpha(q) \le p'$. Then $e_{\alpha+1}(q^\frown\langle\check{c}\rangle) \le p$. At a limit γ, $\mathrm{cf}^W(\gamma) \ge \kappa$ implies $\sup \mathrm{supp}(p) < \gamma$, so induction applies. Otherwise let $\tau = \mathrm{cf}^W(\gamma) < \kappa$, choose a continuous increasing $\langle \gamma_\eta : \eta \le \tau \rangle$ from 0 to γ, and construct

$$q_\eta \in Q_{\gamma_\eta}^V, \qquad e_{\gamma_\eta}(q_\eta) \le p \restriction \gamma_\eta \quad (\eta \le \tau),$$
$$q_\zeta \le q_\eta \quad (\eta < \zeta \le \tau),$$

with omitted coordinates assigned the largest condition. Use $q_0 = 1$ and successor density below $e_{\gamma_\eta}(q_\eta) \cup (p \restriction [\gamma_\eta, \gamma_{\eta+1}))$. At limits $\nu \le \tau$, we have $\langle q_\eta : \eta < \nu \rangle \in V$, since R adds no such sequences; apply closure of the old iteration. Coordinatewise comparison gives $e_\gamma(q_\tau) \le p$. Density and order reflection also give preservation of incompatibility.

Finally, $W \models \kappa^{<\kappa} = \kappa \wedge \sigma = \kappa^+$, so Fact 2.3 gives the σ-chain condition. Each prescribed dense $D_\alpha \subseteq Q_\alpha$ yields a dense $e_\alpha[D_\alpha] \subseteq \widehat{Q}_\alpha$. Under the size hypotheses, $|\delta|, |e_\alpha[D_\alpha]| \le \sigma$ in W. Restricting e_δ below a specified condition gives the same conclusion for its principal ideal. □

5 Names and the bookkeeping function

5.1 Codes for the small iterations

Fix a regular uncountable ρ, put $\sigma = \rho^+$, and assume

$$\rho^{<\rho} = \rho, \qquad 2^\rho = \sigma.$$

Thus $|H_\sigma| = \sigma$ and $\sigma^\rho = \sigma$. The well-founded models used below are coded only in the extensions before forcing at the given level and in extensions by

the specified small complete subforcings, where this arithmetic holds. We make no such cardinality assertion after the full forcing at the lower levels has increased 2^ρ.

A code for a small ordinary iteration consists of a stage wellorder on $J \subseteq \sigma$, names for the orders and linking maps of its iterands on the ordinal σ, enumerations by σ of dense subsets of every initial segment, and the corresponding interpretation of nice names. Unused indices denote the largest condition. The stage wellorder may have order type greater than σ; its cardinality alone is bounded by σ. A component name for an ordinal is coded by at most ρ conditions in deciding antichains. Since a condition has $< \rho$ support, its complete code has hereditary size at most ρ and belongs to H_σ. The iteration code and all the dense-set enumerations together have size σ, and can be coded by classes over H_σ.

Lemma 5.1. *Let M be transitive with $H_\sigma \subseteq M$ and $H_\sigma, \sigma \in M$. Assume M satisfies ZFC without Power Set, with Collection and Choice, the displayed arithmetic, and the stated assertions about codes and forcing. For a code which M regards as a $< \rho$-support iteration of $< \rho$-compact, ρ-linked forcings, write P_α^M, P_α for its internal and external initial segments. Then*

$$\begin{gathered} D \in M,\ M \models D \text{ is dense in } P_\alpha^M \implies D \text{ is dense in } P_\alpha, \\ P_\alpha \text{ is } < \rho\text{-closed and stationary } \sigma\text{-cc}, \\ H_\sigma^{M[G]} = H_\sigma^{V[G]} \quad (G \subseteq P_\alpha \text{ generic}). \end{gathered}$$

Proof. An ill-founded stage order on $J \subseteq \sigma$ would have a descending $s \in {}^\omega J \subseteq H_\sigma \subseteq M$, contrary to M's well-foundedness assertion. Induct through this order. At stage α, the common H_σ gives

$$[\sigma]^{<\rho} \cap M[G_\alpha] = [\sigma]^{<\rho} \cap V[G_\alpha].$$

The iterand order and linking map agree on σ. Finite compatibility is absolute, so every external centered family of size $< \rho$ has its lower bound from $M[G_\alpha]$; equal linking values still give compatibility. Closure and Fact 2.3 apply to the resulting initial segments.

For density, index deciding antichains by the preceding dense-set enumeration. Each has $\leq \rho$ members, so its index-and-value code lies in $H_\sigma \subseteq M$. Complete the antichain with largest-condition values and mix; the resulting name is forced valid by 1. At a limit, code every component this way. For the resulting condition code c_p,

$$|\operatorname{supp}(p)| < \rho, \qquad |\operatorname{tc}(c_p)| \leq \rho, \qquad c_p \in H_\sigma \subseteq M.$$

Its interpretation in M has the same component values; every index below σ denotes an iterand condition. Thus P_α^M is dense in P_α, and every M-dense set remains dense. Only the indexed codes need belong to M.

For $x \in H_\sigma^{V[G]}$, choose a well-founded extensional relation $E \subseteq \gamma^2$ and a point $e < \gamma < \sigma$ with $x = \pi_E(e)$, where π_E is the transitive collapse. A nice name for E uses $\leq \rho$ indexed conditions; its code belongs to the old H_σ, hence to M, and has the same evaluation. The relation is well founded in $M[G]$; Collection supplies its collapse, whose uniqueness gives $x = \pi_E(e) \in M[G]$ with the same hereditary-size bound. Conversely, an internal enumeration witnessing hereditary size $< \sigma$ is also an external one. This proves equality and completes the induction. □

The theory used in the following predicate is fixed as follows. It is ZFC without Power Set, with Collection and Choice, together with the assertions that the distinguished ρ is regular uncountable, $\sigma = \rho^+$, $\rho^{<\rho} = \rho$, and $2^\rho = \sigma$. Its additional first-order assertions say that the displayed stage relation is a wellorder, the coded sequence is its $< \rho$-support iteration, the displayed enumerations are dense, and the iterand names force $< \rho$-compactness and the correctness of their displayed linking maps and bound functions. These are first-order assertions about ordinary set forcing and the displayed data. The required elementary submodels of a sufficiently large H_θ satisfy this theory.

Proposition 5.2. *Under the displayed cardinal arithmetic, there is a first-order predicate* $\mathrm{Cert}(R, \chi)$ *over* H_σ*, with class parameters* R, χ*, such that*

$$R \text{ codes a small iteration} \iff \exists \chi\ \mathrm{Cert}(R, \chi).$$

The predicate uses no parameter for the selected supports or stationary sets.

Proof. Let $\mathcal{T}$ be the fixed theory. The class χ codes $A \subseteq \sigma$, E, Sat, j, and $h, s, r \in A$, with E well founded and extensional and $j : (H_\sigma, \in) \to (A, E)$ an embedding. Require

$$\mathrm{pred}_E(j(a)) = j[a] \quad (a \in H_\sigma),$$
$$\mathrm{pred}_E(h) = j[H_\sigma], \qquad \mathrm{pred}_E(s) = j[\sigma].$$

Compare r and its auxiliary data entry by entry with the input codes. Require the Tarski clauses for Sat and $\forall n \in \omega\, (\mathrm{Ax}_{\mathcal{T}}(n) \Rightarrow \mathrm{Sat}(n))$, including the recursive schemes and the assertions about the coded iteration and its auxiliary data.

All tests are first order over H_σ. In particular, well-foundedness is expressed by

$$\neg\exists f \in H_\sigma\, (f : \omega \to A \ \wedge\ \forall n < \omega\ f(n+1)\ E\ f(n)),$$

since every such sequence of indices belongs to H_σ. If π is the transitive collapse, rank induction gives

$$\pi(j(a)) = a \quad (a \in H_\sigma), \qquad \pi(h) = H_\sigma, \qquad \pi(s) = \sigma.$$

Lemma 5.1 now verifies the iteration properties.

Conversely, take a sufficiently large regular θ and $N \prec H_\theta$ with $|N| = \sigma$, $H_\sigma \subseteq N$, and H_σ, σ, R and its auxiliary data in N. Its collapse fixes H_σ pointwise and hence the iteration code entry by entry. The collapsed model and its satisfaction relation yield χ. □

The iteration code and the well-founded model code can be combined in one class block. A universal quantifier over the forcing class instead ranges over pairs (R, χ) with the first-order antecedent $\mathrm{Cert}(R, \chi)$. Higher class quantifiers remain external; they are not interpreted inside the coded model. This proposition verifies codes for ordinary iterations. The identification of the quotient with such an iteration is proved separately above.

Lemma 5.3. *In W, let R be $< \sigma$-closed and P be σ-cc, using these orders in $R \times P$. Then*

$$H_\sigma^{W[G_P]} = H_\sigma^{W[G_R][G_P]}.$$

The same P-names for iteration and model codes give identical first-order tests.

Proof. Given $r_0 \in R$ and a product name $\dot f : \eta \to \mathrm{Ord}$, $\eta < \sigma$, construct maximal antichains $A_\zeta = \{p_{\zeta,\xi} : \xi < \theta_\zeta\} \subseteq P$ and $r_{\zeta,\xi} \leq r_0$, descending throughout, with $(r_{\zeta,\xi}, p_{\zeta,\xi}) \Vdash \dot f(\zeta) = \check\alpha_{\zeta,\xi}$. While $q \perp A_\zeta$ is possible, extend the current (r, q) to decide; use R-bounds at limits. The chain condition and regularity give $\theta_\zeta < \sigma$ and $\sum_{\zeta<\eta} \theta_\zeta < \sigma$, hence a common lower bound r_*. These antichains define a P-name $\dot g \in W$ with $(r_*, 1_P) \Vdash \dot f = \dot g$. Arbitrary r_0 shows that R adds no $< \sigma$-sequences of ordinals after P. Well-founded relation codes give the equality; common predicates give the consequence. □

For the remainder of this section fix i and put $U_i = \bigcup_{k>i} I_k$. For $C \subseteq I_i$, the closure condition is that $U_i \cup C$ is closed in the full forcing; C itself need not be closed there. In $W = B[G_{D_{>i}}]$ put

$$P_C = (D \restriction (U_i \cup C))/G_{D_{>i}}.$$

In expressions $W[G_C]$, the filter G_C denotes the induced generic filter for this quotient restriction P_C.

5.2 Names over complete subforcings and bookkeeping

Put $\sigma_i = \kappa_{i+1}$. Apply Proposition 4.5 to arbitrary sequences of iterands with the required properties, including restrictions to closed coordinate sets. We first localize the required names and then define the bookkeeping function.

Lemma 5.4. *Let A belong to the ground model and satisfy $|A \cap I_k| \leq \sigma_k$ for every k. Its closure under $\xi \mapsto m(\xi)$ satisfies the same bounds. It has size at most μ and is bounded below Θ.*

Proof. A coordinate of level k can enter the closure only from a coordinate at one of the finitely many levels $1, \dots, k$. There are at most σ_k such starting coordinates, and each contributes at most σ_k level-k coordinates. Each $m(\xi)$ is already closed, so one union suffices. Since Θ is regular and $\mu < \Theta$, the closure is bounded below Θ. If A lies in an initial segment, its closure stays there because dependencies precede their coordinates. □

Proposition 5.5. *Fix i and an initial segment $\alpha \le \Theta$. In $W = B[G_{D_{\alpha,>i}}]$, every ordinal sequence u of length at most σ_i belongs to $B[G_H]$ for a ground-model closed coordinate set $H \subseteq \{\xi < \alpha : i(\xi) > i\}$ with $|H \cap I_k| \le \sigma_k$ for $k > i$. If u enumerates a set C of coordinates, C is contained in a set in the ground model of size at most σ_i.*

Proof. Factor the upper forcing as $R * U_{i+1}$, where $R = D_{\alpha,>i+1}$. By Proposition 4.5, R is $< \kappa_{i+2}$-closed and U_{i+1} is κ_{i+2}-cc, with underlying set of tuples $S_{i+1,\alpha}$ in B.

In $B[G_R]$, a nice U_{i+1}-name for u uses at most σ_i triples consisting of an argument, a condition, and an ordinal value. Replace conditions by their indices in a fixed enumeration of $S_{i+1,\alpha}$ belonging to B. The whole resulting sequence has length at most σ_i. Since $\sigma_i < \kappa_{i+2}$, it belongs to B, although its evaluation need not.

The supports of these σ_i conditions together use at most σ_i level-$(i+1)$ coordinates. Close their union under the sets $m(\xi)$ and apply Lemma 5.4. Each condition, extended by largest conditions on higher coordinates, is a condition in the full forcing. Its membership in the full generic filter is equivalent to membership in the induced G_H. Thus the same name evaluates to u using G_H and $u \in B[G_H]$. The set of all possible ordinal values occurring in the name belongs to B and has size at most σ_i; it contains the range of u. □

Lemma 5.6. *In $W = B[G_{D_{>i}}]$, let $C \subseteq I_i$, $|C| \le \sigma_i$, with $U_i \cup C$ closed; $C \notin B$ is allowed. Let Q be a small Γ_i-forcing over $W[G_C]$, with a chosen $q \in Q$ and finitely many predicate names $\vec{\tau}$. There is a closed $M \in B$ with*

$$\sup M < \Theta, \qquad |M \cap I_k| \le \sigma_k \ (k \ge 1), \qquad C \subseteq K_i = M \cap I_i,$$

and a conditional D_M-name for $(Q, q, \vec{\tau})$, forced to have the required properties and evaluated by the actual generic filter as the old tuple, with the same order on Q.

Proof. Apply Proposition 5.5 to an enumeration of C and close the ground-model cover under $\xi \mapsto m(\xi) \cap I_i$. This fixes $K_i \in B$, $C \subseteq K_i \subseteq I_i$, $|K_i| \le \sigma_i$. Choose $H_0 \subseteq U_i$ in B containing its higher dependencies and the upper coordinates used to localize C. Lemma 5.4 gives the levelwise bounds.

In W, put $P = P_{K_i}$ and choose a dense $E_P \subseteq P$, $|E_P| \le \sigma_i$. Then

$$P \text{ is } \sigma_i\text{-cc}, \qquad P/G_C \text{ is } < \kappa_i\text{-closed and preserves } \sigma_i.$$

The quotient is Γ_i-small. Lemma 4.9 preserves the whole old iteration for Q, its dense-set data and chosen principal ideal. The nice-name count gives $W[G_P] \models 2^{\kappa_i} = \sigma_i$ and $|H_{\sigma_i}^{W[G_P]}| = \sigma_i$. Now choose a fresh certificate χ for the iteration code R_Q by Proposition 5.2, using this actual $H_{\sigma_i}^{W[G_P]}$.

Jointly name $(R_Q, \chi, q, \vec{\tau})$ over P, using condition indices. Canonical P-names for objects of generic H_{σ_i} have $\sigma_i^{\kappa_i} = \sigma_i$ index codes in W. Each class-membership bit and each bit of the embedding of this domain into the coded model is represented by an antichain of size $\leq \kappa_i$. Thus the whole ordinal-index sequence u satisfies

$$|u| \leq \sigma_i \cdot \kappa_i = \sigma_i.$$

Apply Proposition 5.5 to u, enlarge only H_0 and close it, obtaining

$$H \in B, \quad H_0 \subseteq H \subseteq U_i, \quad Z = B[G_H], \quad u \in Z, \quad M = H \cup K_i.$$

Lemma 5.4 gives the bounds and boundedness of M; K_i is unchanged.

For level-i tuples s, t on K_i, the order of $P = D_M/G_H$ obeys

$$s \leq_P^Z t \iff \exists h \in G_H \ (h \cup s \leq_D t) \iff s \leq_P^W t.$$

Restrict an upper witness to H; conversely, amalgamate below h. Every $m(\xi)$ for $\xi \in K_i$ lies in M, so both comparisons use the same component names. The underlying tuples are also the same.

By Proposition 4.5 and Lemma 4.4, in Z,

$$P \text{ is } \sigma_i\text{-cc}, \qquad R = D_{>i}/G_H \text{ is } < \sigma_i\text{-closed}, \qquad W = Z[G_R].$$

The upper quotient R is independent of the level-i generic, so the same product $P \times R$ works in both orders. Lemma 5.3 gives $< \sigma_i$-distributivity of R after P and

$$H_{\sigma_i}^{Z[G_P]} = H_{\sigma_i}^{W[G_P]}.$$

The joint names have identical values; hence their first-order certificate and validity tests transfer to $Z[G_P]$. Proposition 5.2 gives the required properties of Q there. Choose $p \in G_P$ forcing these assertions and those about the chosen condition. Mix the joint name with trivial data off p and regard it as a D_M-name. It is forced valid and has the required actual value and order. □

Remark 5.7. First fix K_i, then preserve Q through the larger level-i subforcing, and only then choose and jointly name the new well-founded model. The subsequent choice of higher coordinates does not enlarge K_i. A model code chosen over the original smaller subforcing may fail to contain the later H_{σ_i}.

Proposition 5.8. *The construction has a bookkeeping function F,* $\operatorname{dom} F = \Theta$*, listing every required name for Q and $Q \restriction q$ ($q \in Q$) over small complete subforcings of the level-i quotients, after enlarging their coordinate sets if necessary. It also lists the term forcings associated with complete subforcings on finitely many lower levels.*

Proof. For closed $M \in B$ with $|M \cap I_k| \le \sigma_k$ ($k \ge 1$), we have $|M| \le \mu$ and $|D_M| \le \mu^\mu = \lambda$. Names represented by maximal antichains use $\le \lambda$ condition–index pairs; the indices code conditions, coordinates $< \Theta$, and ordinals $< \sigma_i$. Thus the collection $\mathcal{B}$ of potential quadruples $(i, M, \dot{Q}, q)$ satisfies

$$|[\Theta]^{\le\mu}| \le \Theta^\mu = \Theta, \qquad |\mathcal{B}| \le \Theta^\lambda = (2^\lambda)^\lambda = \Theta.$$

Choose $F : \Theta \to \mathcal{B}$ with $\sup F^{-1}\{b\} = \Theta$ for every $b \in \mathcal{B}$. At an entry $(i, M, \dot{Q}, q)$, check that M consists of earlier coordinates, is closed, and satisfies the level requirements. Use the conditional name if its deciding condition forces the required properties over the already defined D_M; otherwise use 1.

Expand each coded iteration into its $\le \sigma_i$ constituent iterands at level i, with dependencies on M and earlier coordinates of that block. The cardinal bounds persist. For the block lengths δ_β,

$$1 \le \delta_\beta < \Theta, \qquad \left|\sum_{\eta<\beta} \delta_\eta\right| < \Theta \ (\beta < \Theta), \qquad \sum_{\beta<\Theta} \delta_\beta = \Theta,$$

by regularity of Θ. Every individual iterand therefore has the required properties in a construction of length Θ.

Lemma 5.6 puts any required conditional name over a ground-model closed M with $|M| \le \mu$ and $\sup M < \Theta$. A later occurrence under F supplies it. The enlargement has a Γ_i-small quotient. By Lemma 4.9, the old iteration, its dense-set data and the chosen principal ideal retain the required properties.

For a complete subforcing on levels $< j$, jointly code its order and target name by u, $|u| \le \sigma_j$. A nice name over the level-j quotient uses $\le \sigma_j$ entries. Proposition 5.5 localizes its upper parameters; adjoin the supports of its conditions and close, obtaining

$$H \in B, \quad H \subseteq \bigcup_{k\ge j} I_k, \quad |H \cap I_k| \le \sigma_k \ (k \ge j), \quad \sup H < \Theta.$$

Insert the term forcing after H; the term-forcing theorem gives closure and linkedness. The restriction projection retaining all upper coordinates makes the lower filter generic over the full upper extension containing the term generic filter. Hence the projection from the product to the two-step iteration applies. □

Lemma 5.9. *Fix $j \geq 1$ and work in $W = B[G_{D_{\geq j}}]$. Let C be a retained lower coordinate set with the cardinal bounds of Lemma 7.2, and let P be its quotient with fixed ordinal indexing, $|P| \leq \kappa_j$. If $P \Vdash \dot{E} \subseteq \kappa_{j+1}$, then the joint ordinal-index code of $P, \dot{E}$ has a D_H-name for a closed $H \in B$ satisfying*

$$H \subseteq \bigcup_{k \geq j} I_k, \qquad |H \cap I_k| \leq \sigma_k \ (k \geq j), \qquad \sup H < \Theta.$$

The corresponding term-forcing name can be inserted later at level j, with all dependencies at levels $\geq j$.

Proof. The order of P uses at most κ_j indices. Choose maximal antichains A_α deciding $\alpha \in \dot{E}$, with $|A_\alpha| < \kappa_j$. The joint ordinal-index sequence satisfies

$$u = \mathrm{code}(P, \dot{E}), \qquad |u| \leq \kappa_j + \sum_{\alpha < \sigma_j} |A_\alpha| \leq \sigma_j.$$

Over $B[G_{D_{>j}}]$, a nice name for u over the level-j quotient uses at most σ_j condition indices. Localize this sequence by Proposition 5.5, adjoin the supports of those conditions, and close under dependencies using Lemma 5.4. Their level-j supports use at most σ_j coordinates. This gives the required H, with $u \in Z = B[G_H]$ and $|H| \leq \mu$; regularity of Θ gives $\sup H < \Theta$.

Lower coordinates occur in u as ground-model ordinal parameters, not as generic filters. An inserted block may therefore use

$$m(\xi) = H \cup J_\xi \subseteq \bigcup_{k \geq j} I_k,$$

where J_ξ consists of earlier coordinates of that block. Since C is bounded, choose the block above $\sup(H \cup C)$.

In Z, use the same ordinal-indexed conditions and order of P. The enumeration preserves the size bound, and its chain condition follows downward from W. Theorem 2.9 applies to

$$T = \mathrm{Term}(P, \mathbb{A}_{\kappa_j}(\dot{E}))^Z.$$

Subsequent forcing uses this old T with its fixed conditions and order, including after the lower generic filter is added.

Before the recursion, list cofinally often all potential names of size $\leq \lambda$ over potential bounded coordinate sets with the displayed bounds. There are at most $\Theta^\lambda = \Theta$ such syntactic ordinal-index codes. At each stage check coordinates and forcing assertions only over the already constructed restriction. Every actual code passes at a later occurrence because H is bounded. This uses the structural and localization lemmas for arbitrary partial constructions, without assuming universality of the final bookkeeping function. □

5.3 Properties of the forcing

The choices can be made uniformly in $\vec{T}$ over L. Represent the potential ordinal-index codes by $C_{\vec{T}}$-names and use the $<_L$-least enumerations of their antichain codes. Since $|C_{\vec{T}}| \leq \lambda$ and $\Theta^\lambda = \Theta$ in L, a name for any of the required ordinal-index codes has a representation with at most λ entries, and there are at most Θ such representations. At each step the validity check is a forcing assertion over the already defined restriction. The same definable recursion therefore produces a $C_{\vec{T}}$-name $\dot{D}_{\vec{T}}$ for every $\vec{T} \in L$. This uses only ordinary set-forcing names and the constructible wellorder.

Theorem 5.10. *For each $B = B_{\vec{T}}$ there is a forcing D with the following properties. Its bookkeeping function lists all ordinary names coded by ordinals over ground-model closed sets of coordinates satisfying the bounds above, as specified in Proposition 5.8. The recursion is uniform in B.*

(M1) for every $j \geq 1$, $D_{\geq j}$ is a complete subforcing, the restriction map has the extension property stated in Section 2, and $D_{\geq j}$ is $< \kappa_j$-closed;

(M2) in $B^{D_{>i}}$, the quotient $D_{\geq i}/G_{D_{>i}}$ is forcing equivalent to a $< \kappa_i$-support iteration with the stationary κ_{i+1}-chain condition. Every class over H_{κ_i} in the extension by this quotient belongs to the extension by a complete subforcing obtained from at most κ_{i+1} coordinates closed under the remaining dependencies. Any previously chosen such set can be retained in this enlargement. Quotients between these subforcings are small Γ_i-iterations. Every small Γ_i-forcing and every restriction of it below a condition, named over a retained subforcing, occurs over that subforcing or over a small enlargement using the canonical lift of the old name. This applies also to sets of coordinates which are not initial segments;

(M3) every finite family of classes over HC *in B^D belongs to $B[G_C]$ for a complete subforcing D_C, where C is closed and $|C \cap I_i| \leq \kappa_{i+1}$ for every i. The levels meeting C may be unbounded;*

(M4) D has a dense subset of cardinality at most Θ and is Θ-cc. It preserves all cardinals and adds no reals;

(M5) small Γ_i-iterations admit uniform codes using ordinal indices. Such a code and a code for a well-founded model witnessing that it defines an iteration of the required kind are classes over $H_{\kappa_{i+1}}$ whose correctness is expressed by a first-order formula. This formula characterizes exactly the required iterations before forcing at level i and in the extensions by the small complete subforcings, where $2^{\kappa_i} = \sigma_i$. It uses neither the sets of selected labels nor the added clubs as parameters. The Boolean-valued translation of names uses these codes, with

the remaining class quantifiers ranging over all classes in the ambient universe;

(M6) for every relative complete subforcing on finitely many lower levels and every target name in Lemma 7.2, the corresponding term forcing occurs after all coordinates on which its name depends. The evaluated lower filter is generic over the full upper extension containing the generic filter for the term forcing.

Proof. At each stage, a name is checked only over an already defined forcing D_M. Proposition 4.5 applies to every such sequence of iterands with the required properties, regardless of whether it is universal. Its assertions about initial segments therefore hold throughout the recursion. The choice of complete subforcings and Lemma 5.6 can then be applied to the resulting sequence; they prove the required bookkeeping property.

A class over HC has a nice name over the level-one quotient with κ_1 entries. Its ground-model index code and the closure of its support give a ground-model closed coordinate set satisfying the bounds at each level and yielding a complete subforcing. The same argument for each quotient includes the finite families of parameters and enlarged complete subforcings used in the computation of truth. Proposition 5.8 supplies the generic filters for the prescribed principal ideals over these subforcings and the term forcings for successive almost-disjoint codings. Proposition 5.2 gives the first-order tests for the small forcing class used in the reduction of complexity.

Closure of the upper forcings and the chain conditions of the level-i quotients follow from the quotient theorem. Lemmas 4.7 and 4.8 give the size bound, the Θ-chain condition, preservation of all cardinals, and the absence of new reals. □

5.4 Names over arbitrary complete subforcings

Lemma 5.11. *Work in $B^{D_{>i}}$ and let $C \subseteq I_i$ have cardinality at most κ_{i+1}, with $U_i \cup C$ closed as above. In this lemma write $P_C = (D \restriction (U_i \cup C))/G_{D_{>i}}$. Factor the quotient forcing through P_C first.*

(a) Every class over H_{κ_i} in the full level-i extension belongs to the extension by $P_{C'}$ for some $C' \supseteq C$ with $|C'| \leq \kappa_{i+1}$ and $U_i \cup C'$ closed. The quotient $P_{C'}/G_{P_C}$ is a small Γ_i-iteration.

(b) Every name over the extension by P_C for a small Γ_i-forcing, or for its restriction below a condition, is handled at a later coordinate whose set of dependencies contains C and the ground-model coordinates required for that name.

Proof. For (a), use a nice name over the full level-i quotient. The set H_{κ_i} has cardinality κ_i, and every deciding antichain has cardinality at most κ_i. Thus the name uses at most κ_i conditions. Adjoin their level-i supports to C and close under the remaining dependency sets. Lemma 5.4 bounds this closure by κ_{i+1}. The resulting complete restriction contains the class.

To identify its quotient over C, order the retained coordinates of C first and then the other coordinates in their original order. Closedness ensures that this ordering puts every dependency before the coordinate which uses it. For each new factor, the extra preceding retained coordinates give a $< \kappa_i$-closed quotient. The old factor therefore retains compactness and its linking map. The proof of Proposition 4.5, with the prescribed restriction on C kept in its generic filter, identifies the quotient with the corresponding $< \kappa_i$-support iteration. It has at most κ_{i+1} stages and a dense subset of that cardinality, so it is small. If a ground-model covering set is needed, apply Proposition 5.5 to an enumeration of C and use the closure construction of Lemma 5.6.

For (b), apply Lemma 5.6 jointly to the forcing name, its chosen condition, and its predicate names. It gives a bounded ground-model closed set M, whose level-i part K contains C, and a conditional name over D_M. The quotient from C to K is small by (a). Lemma 4.9 shows that the old forcing and the specified principal ideal retain their required properties after this enlargement. The deciding condition for the conditional name belongs to the actual generic filter.

Proposition 5.8 therefore inserts the specified principal ideal at an occurrence after every coordinate in M. In the level-i iteration over the full higher extension, its generic filter is generic over the entire preceding initial segment, and hence over the retained extension by K. The old and interpreted forcing orders agree by Lemma 4.9. Thus this is a generic filter for precisely the requested forcing below the requested condition. Adjoining the inserted block and closing its dependencies still uses at most κ_{i+1} level-i coordinates. This also gives a small retained extension containing its generic filter and the evaluations of the specified names. □

6 Forcing and the reduction of complexity

6.1 The small forcing class

Fix a level i, and write $\rho = \kappa_i$ and $\sigma = \rho^+$. All models in which the small forcing class is used satisfy

$$\rho^{<\rho} = \rho, \qquad 2^\rho = \sigma.$$

Here a small Γ_i-forcing is a forcing with its specified code for a $< \rho$-support iteration of cardinality at most σ, whose iterands are $< \rho$-compact and ρ-linked and have dense subsets indexed by σ. We also include restrictions

below conditions and forcings with specified dense equivalences to these iterations. This is the class defined in Section 4; no selected stationary set or support is part of its definition.

Lemma 6.1. *Every small Γ_i-forcing is $< \rho$-closed, is σ-cc, and has a dense subset of cardinality at most σ. It preserves the displayed cardinal arithmetic. The small forcing class has the following closure properties.*

1. *If P, Q are small forcings in N, then in a P-extension the old order Q is forcing equivalent to the iteration obtained by interpreting its specified iterand names there. In particular it remains a small Γ_i-forcing. The product $P \times Q$ is forcing equivalent to a small Γ_i-forcing.*

2. *If P is small and $P \Vdash \dot{Q} \in \Gamma_i$ is small, then $P * \dot{Q}$ is forcing equivalent to a small Γ_i-forcing.*

3. *The same assertions hold after restriction below any specified condition.*

Proof. Closure follows from the usual lower-bound construction for $< \rho$-support iterations. Compactness gives countable parallel closure, and linkedness gives the stationary ρ^+-chain condition. Fact 2.3 therefore gives the σ-chain condition of every initial segment and of the whole iteration. Using these chain conditions, represent component names by maximal antichains of cardinality at most ρ. Inductively a condition represented this way has an index code of cardinality at most ρ, using only indices below σ. Since $\sigma^\rho = \sigma$, the iteration has a dense subset indexed by σ.

The iteration adds no sequences of ordinals of length less than ρ. It preserves σ, and a nice name for a subset of ρ has at most ρ indexed entries. There are at most $\sigma^\rho = \sigma$ such names. Thus it preserves both equations in the displayed cardinal arithmetic.

Let $P, Q \in N$ be small. The first part of the proof shows that P is $< \rho$-closed and preserves σ. Apply Lemma 4.9 with $R = P$ to the specified iteration for Q. Its old dense order remains dense in the interpreted iteration, with the same order and incompatibility relation. Its iterands remain $< \rho$-compact and ρ-linked, and its prescribed dense subsets still have size at most σ. Thus the old Q remains small in the P-extension, as does each specified principal ideal.

Concatenate the iteration for P with this interpretation of Q to obtain (1). Concatenation also proves (2). If the stage order of $\dot{Q}$ is named rather than fixed, choose a bound below σ^+ for its order type using the σ-chain condition, and pad by trivial iterands to this bound. The resulting stage order still has cardinality at most σ. Restricting the individual factors below the components of a condition, and using the corresponding conditional names, gives (3). Compactness and linkedness are preserved by these restrictions. □

6.2 Names and second-order forcing

We record precisely the interpretation of class quantifiers in forcing statements. All such quantifiers below range over classes of the ambient universe, rather than over classes belonging to a model used to verify a forcing code.

Lemma 6.2. *Let $\tau > \omega$ be regular, $\tau^{<\tau} = \tau$, and let Q be τ-cc with a specified dense order indexed by τ. For each fixed finite second-order formula over $H_\tau^{N[G_Q]}$, its forcing relation has a uniform translation over H_τ^N with the same class-quantifier prefix. In particular,*

$$\textit{first order} \longmapsto \textit{first order}, \qquad \Sigma^2_m \longmapsto \Sigma^2_m, \qquad \Pi^2_m \longmapsto \Pi^2_m.$$

Its parameters are the class codes of the dense order and its predicate names.

Proof. Put $H = H_\tau^N$ and use the indexed dense preorder, with unused indices denoting 1. Represent a pointed relation on a subset of $\gamma < \tau$ by nice names for its domain and relation, and a maximal antichain deciding its distinguished point. There are $< \tau$ atomic bits, each decided by an antichain of size $< \tau$; the indexed code lies in H. Every generic H_τ object has such a well-founded extensional relation name: the chain condition bounds γ below a given condition p, and mixing with the empty set off p gives a globally valid name.

Write $\mathrm{Obj}(a)$ for validity of such a code, representing x_a. Before testing well-foundedness, validate direct graph names for $f : \omega \to \gamma$ without using object-collapse codes. Every descending-sequence witness has such a code in H. Equality and membership are tested by names for isomorphisms between the appropriate pointed subrelations. The maximum principle gives direct graph names with $< \tau$ entries; conversely, an isomorphism identifies the collapses. These tests, and hence the atomic Boolean values, are first order over H.

Represent Boolean values by antichain codes $A \in H$, modulo equality of their values $[A]$. Their order is first order:

$$[A] \le [B] \quad \iff \quad \forall a \in A\, \forall q \le a\, \exists r \le q\, \exists b \in B\ (r \le b).$$

Equality and complement are consequently first order. Joins of class-specified families are their least upper bounds in this order; all quantified bounds are still antichain codes in H. Existence follows from the chain condition and the ordinary Boolean completion. No definable choice of representatives or extra class quantifier is required.

A class-name table χ assigns a Boolean value to each valid object code, subject to

$$\|x_a = x_b\| \wedge \chi(a) = \|x_a = x_b\| \wedge \chi(b) \quad (\mathrm{Obj}(a), \mathrm{Obj}(b)).$$

Name(χ) asserts this total assignment modulo Boolean equality and the displayed coherence; it is first order. Every ambient predicate name gives such a class over H; every coherent table defines a predicate name whose interpretation is independent of the chosen object code.

For a fixed first-order formula θ, use Boolean operations and joins or meets over Obj(a) at object quantifiers. Finite recursive substitution in its syntax defines $\|\theta\|$ first order, and $p \Vdash \theta$ means $[p] \leq \|\theta\|$. No simultaneous satisfaction or valuation class is quantified.

Finally, the maximum principle and mixing with the empty predicate off p give

$$p \Vdash \exists X\, \psi(X) \iff \exists \chi\, [\mathrm{Name}(\chi) \wedge p \Vdash \psi(\chi)],$$
$$p \Vdash \forall X\, \psi(X) \iff \forall \chi\, [\mathrm{Name}(\chi) \Rightarrow p \Vdash \psi(\chi)].$$

Here χ ranges over classes over H in the ambient universe. Recursion along the finite prefix preserves its exact class-block pattern. □

In the application of this lemma, $\tau = \sigma$. The hypotheses hold because $2^\rho = \sigma$ implies $\sigma^{<\sigma} = \sigma$. The small forcing is σ-cc and has density at most σ. A name for a subset of H_ρ has a code which is an *element* of H_σ: its hereditary size is at most ρ. By contrast, a name for an arbitrary class over the generic H_σ is represented by a *class* table. This distinction is used at every step of the reduction of complexity.

6.3 The transformation T_i

We use prenex second-order formulas and combine the finitely many variables in each class block into one class variable by the fixed pairing functions. First-order formulas are included as rank zero. The recursion stops at a first-order formula or at a single class block followed by a first-order formula. The clauses for a longer prefix are then applied only outside this terminal block.

Definition 6.3. *Work in a model N of the displayed cardinal arithmetic, and let the class parameters belong to N. On a terminal first-order, $\Sigma^2_1(H_\rho)$, or $\Pi^2_1(H_\rho)$ formula, let $T_i(\varphi)$ be its first-order translation over H_σ: relativize object quantifiers to the definable set H_ρ, and replace class quantifiers over H_ρ by object quantifiers over its subsets, all of which belong to H_σ. Class parameters are interpreted as these same objects. For longer prefixes put*

$$T_i(\neg\varphi) = \neg T_i(\varphi).$$

If $\exists X\, \varphi$ has at least two class blocks, let $T_i(\exists X\, \varphi)$ assert that there are a small $Q \in \Gamma_i$, a condition $q \in Q$, and a Q-name $\dot{X}$ for a subset of H_ρ, such that

$$q \Vdash_Q T_i(\varphi)(\dot{X}).$$

The universal clause is obtained by negation. The forcing relation in this definition is the ordinary set-forcing relation for the already constructed shorter formula, interpreted second order in the ambient universe.

Lemma 6.4. *For small $P \in \Gamma_i$, old parameters, and any model N of the displayed arithmetic,*

$$N \models T_i(\varphi) \quad \iff \quad N[G_P] \models T_i(\varphi).$$

Proof. Induct on the nonterminal class blocks. Closure and Fact 2.5 give the terminal case and fix H_ρ; negation is immediate.

At an existential step, put $\psi = T_i(\varphi)$. For old witnesses $Q, q, \dot{X}$, Lemma 6.1 keeps each factor of $P \times Q$ small after the other. For every Q-generic $G_Q \ni q$ over $N[G_P]$, product genericity and induction give

$$N[G_Q] \models \psi(X) \implies N[G_P][G_Q] \models \psi(X), \qquad X = \dot{X}^{G_Q}.$$

Hence $N[G_P] \models q \Vdash_Q \psi(\dot{X})$, with the old Q still small.

Conversely, name witnesses from $N[G_P]$ by $\dot{Q}, \dot{q}, \ddot{X}$, and choose $p \in G_P$ forcing their required properties. Lemma 6.1 and the two-step forcing theorem give

$$S = (P \restriction p) * (\dot{Q} \restriction \dot{q}) \in \Gamma_i, \qquad 1_S \Vdash_S \psi(\dot{X}^*),$$

where $\dot{X}^*$ is the composite name. Thus $S, 1_S, \dot{X}^*$ witness $T_i(\exists X\, \varphi)$ already in N. □

Lemma 6.5. *Put $V_i = B[G_{D_{\geq i}}]$, $W = B[G_{D_{>i}}]$ and $N_C = W[G_C]$, where C is small and retained at level i. Then*

$$V_i \models \varphi(\vec{a}) \quad \iff \quad N_C \models T_i(\varphi)(\vec{a}) \quad (\vec{a} \in N_C).$$

Proof. The structural results give a $< \rho$-closed extension $N_C \subseteq V_i$ and, for every $X \subseteq H_\rho$ in V_i, a small retained $C' \supseteq C$ with $X \in N_{C'}$ and a small quotient over N_C. A prescribed $Q \restriction q$ is inserted after some small $C_0 \supseteq C$, generically over N_{C_0}. These are Lemma 5.11, including its assertion for specified principal ideals.

Induct on the nonterminal prefix, simultaneously for all retained sets and their parameters. Closure and Fact 2.5 handle terminal blocks; negation follows by induction.

If $V_i \models \exists X \varphi(X, \vec{a})$, localize a witness to $N_{C'}$. Induction gives $N_{C'} \models T_i(\varphi)(X, \vec{a})$. For its small quotient R over N_C, the forcing theorem gives $q \in G_R$ and $\dot{X}$ with $q \Vdash_R T_i(\varphi)(\dot{X}, \vec{a})$. Thus $(R, q, \dot{X})$ witnesses $N_C \models T_i(\exists X \varphi)(\vec{a})$.

Conversely, choose witnesses $Q, q, \dot{X} \in N_C$ and enlarge to C_0 for their insertion. The product argument of Lemma 6.4, using invariance of the shorter formula, gives $N_{C_0} \models q \Vdash_Q T_i(\varphi)(\dot{X}, \vec{a})$. The inserted $Q \restriction q$ supplies a Q-generic H over N_{C_0} with $q \in H$. Choose small retained $C' \supseteq C_0$

containing this block and its dependencies, with $N_{C_0}[H] \subseteq N_{C'}$ and the remaining quotient small. Then, with $X = \dot{X}^H$,

$$\begin{aligned} N_{C_0}[H] \models T_i(\varphi)(X, \vec{a}) &\Longrightarrow N_{C'} \models T_i(\varphi)(X, \vec{a}) \\ &\Longrightarrow V_i \models \varphi(X, \vec{a}), \end{aligned}$$

by Lemma 6.4 and induction. Hence $V_i \models \exists X \varphi(X, \vec{a})$. □

Lemma 6.6. *For $k \geq 2$ the transformation can be chosen uniformly so that*

$$\begin{aligned} T_i(\Sigma^2_k(H_{\kappa_i})) &\subseteq \Sigma^2_{k-1}(H_{\kappa_{i+1}}), \\ T_i(\Pi^2_k(H_{\kappa_i})) &\subseteq \Pi^2_{k-1}(H_{\kappa_{i+1}}). \end{aligned}$$

It introduces no parameters for the selected supports, clubs, or bookkeeping function.

Proof. Since $\mathcal{P}(H_\rho) \subseteq H_\sigma$ and H_ρ, ρ are definable in H_σ, terminal first-order and $\Sigma^2_1/\Pi^2_1(H_\rho)$ formulas translate first order over H_σ, with their old class parameters as objects.

For $\exists X \varphi \in \Sigma^2_k(H_\rho)$, use one class witness Z, with fixed pairing and sections

$$Z_0 = R, \qquad Z_1 = \chi, \qquad Z_2 = \{q\}, \qquad Z_3 = \{a\},$$

where $R, \chi \subseteq H_\sigma$ code Q and its witnessing model, $q, a \in H_\sigma$, and a codes the name $\dot{X}$. Proposition 5.2 makes $\mathrm{Cert}(R, \chi)$ first order. The tests on q, a are also first order. Their uniquely designated values are independent of later classes, so their first-order recovering quantifiers may follow the remaining class prefix.

By induction and Lemma 6.2, $q \Vdash_Q T_i(\varphi)(\dot{X})$ is first order over H_σ for $k = 2$, and $\Pi^2_{k-2}(H_\sigma)$ for $k > 2$. The single leading $\exists Z$ block therefore yields $\Sigma^2_{k-1}(H_\sigma)$ in both cases. Negation gives the Π^2_{k-1} assertion.

All tests use the actual ambient forcing class. Coded models are used only for Cert; every remaining class quantifier ranges over ambient classes. Finite recursion on the original formula code, using the definable cardinals and forcing classes, proves uniformity and the asserted parameter dependence. □

6.4 Reduction to the club-forcing extension

Proposition 6.7. *For every Π^2_n formula A on the reals there is a uniformly obtained $\Pi^2_1(H_{\kappa_n})$ formula Φ_A, independent of $\vec{T}$, such that*

$$B_{\vec{T}}[G_{D_{\vec{T}}}] \models A(a) \quad \Longleftrightarrow \quad B_{\vec{T}} \models \Phi_A(a). \tag{1}$$

The corresponding statement holds for Σ^2_n formulas, with $\Sigma^2_1(H_{\kappa_n})$ in place of $\Pi^2_1(H_{\kappa_n})$. Only the original real parameters occur.

Proof. Apply the truth and complexity lemmas successively for $i = 1, \dots, n-1$, with the empty retained level-i set each time. The successive ambient models are $B_{\vec{T}}[G_{D_{\geq i+1}}]$; they satisfy the cardinal arithmetic required at level i because the upper forcing is $< \kappa_{i+1}$-closed. At each step the complexity drops by one and the first-order domain increases from H_{κ_i} to $H_{\kappa_{i+1}}$.

The resulting assertion is $\Pi^2_1(H_{\kappa_n})$ in $B_{\vec{T}}[G_{D_{\geq n}}]$. The forcing $D_{\geq n}$ is $< \kappa_n$-closed, so Fact 2.5 transfers it to $B_{\vec{T}}$. For $n = 1$ use this last argument directly. The same reasoning treats the existential dual. The forcing classes and their tests have no selected-support parameters, and all operations on formula codes are uniform finite syntactic recursions. Hence the resulting formula has precisely the asserted parameter dependence. □

7 The coding predicates

7.1 The first level

Put

$$\mathrm{Code}_1(t) \iff \exists C \subseteq \kappa_1 \, [C \text{ is club and } C \cap S^1_t = \varnothing].$$

By Proposition 3.5(S2), Code_1 is Σ^2_1 over HC.

7.2 Successive almost disjoint coding

Write $Z_1(X) = X$ and

$$Z_{r+1}(X) = \{\eta < \kappa_{r+1} : Z_r(X) \cap A^r_\eta \text{ is bounded in } \kappa_r\}.$$

These recursively decoded sets belong to every relevant intermediate extension containing X and the canonical almost disjoint families. Their values are absolute to further extensions.

Lemma 7.1. *Let $V \subseteq W$ be transitive models, let $P \in V$ have the same conditions and order in W, and put $T = \mathrm{Term}^V(P, \mathbb{A}_\rho(\dot{E}))$ for a P-name $\dot{E} \in V$. Suppose $H \in W$ is T-generic over V and G_P is P-generic over W. Then (G_P, H) is $P \times T$-generic over V and $W[G_P]$ contains a subset of ρ which almost disjointly codes $\dot{E}[G_P]$.*

Proof. Since $V[H] \subseteq W$, the filter G_P is P-generic over $V[H]$. The iteration $T * \check{P}$ is the product $T \times P$. Apply the projection $P \times T \longrightarrow P * \mathbb{A}_\rho(\dot{E})$ from Theorem 2.9. The generic filter induced by this projection gives the required almost disjoint code. Boundedness of the intersection of two fixed sets in a fixed ordinal is absolute to further forcing extensions. □

Lemma 7.2. *Assume Theorem 5.10. Fix $j \geq 1$ and work in $W = B[G_{D_{\geq j}}]$. Let C consist of coordinates of levels below j, with at most κ_{k+1} coordinates*

at each level $k < j$. Suppose that its union with all coordinates of level at least j is closed under the sets $m(\xi)$, and put

$$P = \big(D \restriction (\{\xi : i(\xi) \geq j\} \cup C)\big)/G_{D_{\geq j}}.$$

Then P has a dense subset of cardinality at most κ_j and satisfies the κ_j-chain condition. At $j = 1$ it is trivial.

If $\dot{E}$ is a P-name for a subset of κ_{j+1}, the bookkeeping function selects $\mathrm{Term}(P, \mathbb{A}_{\kappa_j}(\dot{E}))$ at a level-j stage after all coordinates needed for the code of P and the name $\dot{E}$. Its generic filter and the induced P-generic filter give a generic filter for $P * \mathbb{A}_{\kappa_j}(\dot{E})$.

Proof. There are finitely many levels below j. Reorder them in descending order. The bounds on their coordinate sets and the ordinary iteration description of each quotient give the stated density and chain condition. This argument is in W, where the higher generic filters have already been fixed. Before the stage at which the term forcing is used, $\kappa_j^{<\kappa_j} = \kappa_j$ and $\kappa_{j+1}^{<\kappa_j} = \kappa_{j+1}$: neither the higher forcing nor an initial segment of the level-j quotient adds shorter sequences of ordinals. Theorem 2.9 therefore shows that this term forcing has the required closure, linking, and size properties.

Propositions 5.5 and 5.8 ensure that it is used with these particular parameters. Let H be its generic filter. The induced lower filter is P-generic over W, which contains H, by completeness of the restriction retaining all higher coordinates. The two filters are thus product generic over the model in which the term forcing was defined. Apply the projection of Theorem 2.9. Completeness of a restriction without first retaining the higher coordinates would not by itself give this product genericity. □

Proposition 7.3. *Assume Propositions 4.5, 5.5, and 5.8. Fix $j \geq 1$, put $W_j = B[G_{D_{\geq j}}]$, and let*

$$C \subseteq \bigcup_{1 \leq k < j} I_k, \qquad |C \cap I_k| \leq \kappa_{k+1} \quad (1 \leq k < j),$$

with $C \cup \bigcup_{k \geq j} I_k$ closed. Put

$$Q_j = \big(D \restriction (C \cup \bigcup_{k \geq j} I_k)\big)/G_{D_{\geq j}}.$$

For every $E \subseteq \kappa_{j+1}$ in $W_j[G_{Q_j}]$, the final model contains $X \subseteq \kappa_1$ with $Z_{j+1}(X) = E$.

Proof. For $1 \leq r \leq j$, put $C_r = C \cap \bigcup_{1 \leq k < r} I_k$, and define W_r, Q_r as above. Dependencies never decrease the level, so $C_r \cup \bigcup_{k \geq r} I_k$ is closed. By Proposition 4.5, choose an ordinal-indexed P_r forcing equivalent to Q_r, with

$$|P_r| \leq \kappa_r, \qquad P_r \text{ is } \kappa_r\text{-cc}; \qquad Q_1 = 1.$$

Starting with $E_{j+1} = E$, construct downward sets $E_r \in W_r[G_{Q_r}]$, $E_r \subseteq \kappa_r$, satisfying

$$E_{r+1} = \{\eta < \kappa_{r+1} : E_r \cap A^r_\eta \text{ is bounded in } \kappa_r\}.$$

At step r, a nice P_r-name $\dot{E}_{r+1}$ has $\leq \kappa_{r+1}$ indexed entries, since its deciding antichains have size $< \kappa_r$. Localization puts $P_r, \dot{E}_{r+1}$ in an earlier upper extension $V_r \subseteq W_r$, with the same ordinal order. The bookkeeping function later selects $T_r = \mathrm{Term}^{V_r}(P_r, \mathbb{A}_{\kappa_r}(\dot{E}_{r+1}))$ at level r. Completeness of the restriction retaining all upper coordinates makes the induced lower filter generic over W_r. Transfer it to P_r in W_r, which contains the term generic filter. Lemma 7.1 gives the required E_r. The chosen dense embedding need not belong to V_r.

For $r > 1$,

$$W_r[G_{Q_r}] \subseteq W_{r-1}[G_{Q_{r-1}}].$$

Indeed, W_{r-1} contains the retained level-$(r-1)$ generic and all preceding term generics. Thus the same procedure applies to E_r over C_{r-1}, retaining no additional lower coordinates. Choose each bookkeeping occurrence after all previously used stages. At $r = 1$ the lower quotient is trivial, so almost disjoint coding applies directly. With $X = E_1$, the defining recursion gives $Z_r(X) = E_r$ for $1 \leq r \leq j+1$, hence $Z_{j+1}(X) = E$. □

Lemma 7.4. *If $t \notin T_n$, the set S^n_t remains stationary in $B_{\vec{T}}[G_{D_{\vec{T}}}]$.*

Proof. It is stationary in $B_{\vec{T}}$ by Lemma 3.3. First force with $D_{\geq n}$, which is $< \kappa_n$-closed. The assertion that there is a club disjoint from this fixed stationary set is existential second order over H_{κ_n}. Fact 2.5 shows that no such club is added.

The remaining finitely many lower quotients form a κ_n-cc iteration. Such forcing preserves stationary subsets of κ_n. To see this directly, given a name for a club, for each $\alpha < \kappa_n$ take a maximal antichain deciding its first point above α. There are fewer than κ_n possible values; bound them by $f(\alpha) < \kappa_n$. The ground-model club of limit ordinals closed under f is forced to be contained in the named club. It meets S^n_t. For $n = 1$ the lower quotient is trivial. □

7.3 The coding definitions

We make explicit the coding conventions used in this subsection. At $\rho = \kappa_r$, take A^r_η to be the very same canonical family denoted by B_η in Section 3.1. We use the following consequences of the forcing construction:

(C1) The cardinals are preserved, and for every finite family of classes over HC there is a complete restriction satisfying the levelwise cardinal bounds whose extension contains that family.

(C2) If $r \geq 2$ and $E \subseteq \kappa_r$ belongs to an extension by a complete restriction on the levels below $r-1$ over $B[G_{D_{\geq r-1}}]$, then there is a class $X \subseteq \kappa_1$ with $Z_r(X) = E$.

(C3) Each stationary set S_t^n with $t \notin T_n$ is stationary in the final extension. For $t \in T_n$ the final extension contains an avoiding club and its padded code from Lemma 3.1.

The first assertion is the localization assertion in Theorem 5.10, and the second is Proposition 7.3. The stationary-set assertion is Lemma 7.4; the selected padded targets already belong to $B_{\vec{T}}$. All representation assertions below use only these particular intermediate extensions.

Fix once and for all constructible coding functions for finite tuples of ordinals, finite sequences, and finite formal terms. Their restrictions to an infinite cardinal τ take values below τ. We use disjoint columns for equality, membership, and satisfaction. An ordinal code similarly has separate columns for its domain and its strict order, so that the empty ordinal is included. The language before adding constants is countable and recursive. It includes symbols for the Skolem functions of every existential formula, and, for each $j < r$, symbols F_j, H_j. The formal constants are

$$\{c_\alpha, g_\alpha : \alpha < \kappa_r\} \cup \{b_0, \dots, b_{r+2}, z_1, \dots, z_k, t_1, \dots, t_l, d_1, \dots, d_m\}.$$

Here the g_α enumerate generators, c_α are ordinal constants, the b_j are designated cardinals, the z_i are designated decoded sets, and the t_i are real parameters. The d_i are finitely many auxiliary distinguished objects, such as the order types and isomorphisms in a witness. Formula numbers are natural numbers. A satisfaction entry consists of a formula number and a finite assignment of formal terms; the uncountably many constants are thus treated as assignments, rather than as part of the recursive enumeration of formulas.

The coding of these columns is fixed in advance. Write $q_{\mathrm{eq}}(u, v)$, $q_{\mathrm{mem}}(u, v)$ and $q_{\mathrm{sat}}(e, \vec{u})$ for their ordinal indices. In particular, these indices are independent of the model being coded. The value of a constructor for a formal term is another fixed ordinal function of its finitely many arguments.

For $0 < a < \kappa_{j+1}$ let $f_{j,a}$ be the $<_L$-least surjection from κ_j onto a. The exact-initial-segment definition of $<_L$ in Section 3.1 gives functional Σ_1 definitions of these graphs, uniformly in j. Indeed, supply a candidate surjection and its exact $<_L$-initial segment, and check that no preceding candidate is a surjection with the same domain and range. Use the same method for the canonical coding functions. The suitable theories below require the relevant graphs to be total and functional over their displayed L_β on the specified domains. Upward Σ_1 absoluteness then identifies their values with the actual constructible values.

Definition 7.5. *For each $r \geq 1$ let T_r consist of ZFC without Power Set, with Collection and Choice, together with the following axioms in the indicated Skolem language:*

(i) $b_j = \omega_j$ *for* $0 \leq j \leq r+2$, *and* $b_{r+2} = \omega^L_{r+2}$. *Put* $\rho = b_r$ *and* $\beta = b_{r+1}$. *The set* L_β *exists.*

(ii) The canonical diamond and almost-disjoint graphs at ρ *are total and functional over* L_β *on exactly the domains specified in Section 3.5. The same is required of the fixed canonical coding functions and their inverses; their restrictions at* β *satisfy the corresponding restriction identities for the even and odd parts. For each* $j \leq r$, *the canonical graph of* $a \mapsto f_{j,a}$, *evaluated over this same* L_β, *is total and functional for* $0 < a < b_{j+1}$.

(iii) Every z_i *is a subset of* ρ, *and every* t_i *is a subset of* ω.

(iv) For $j < r$ *and* $0 < a < b_{j+1}$, $F_j(a)$ *is a function from* b_j *onto* a, *and, for* $u \in a$,

$$H_j(a,u) \in b_j, \qquad F_j(a)(H_j(a,u)) = u.$$

The finitely many displayed axioms and the recursive axiom schemes are checked using their natural-number formula codes. Skolem functions are additional interpretations on the coded universe; they need not be elements of that universe.

For the representation direction we use transitive models of T_r of cardinality κ_r with true $b_j = \kappa_j$ for $j \leq r$. Such models are supplied by the transitive collapses of the elementary submodels used below. For the soundness direction, the equality of the designated lower cardinals with the true cardinals will be a conclusion of the coding clauses, not an additional external assumption on a proposed code.

Lemma 7.6. *There is a first-order predicate* $V_1(Y; X_1, \ldots, X_k, t_1, \ldots, t_l)$ *over* HC *which says that* Y *directly codes a well-founded extensional model* N *of* T_1, *its satisfaction relation and its Skolem functions, with true* $b_1 = \omega_1$, *with* $z_i = X_i$, *and with the indicated real parameters. Every such model of cardinality* ω_1 *has a direct code.*

Proof. The relation, its domain and its satisfaction relation are coded directly by classes on ω_1. Their finite structural and Tarski clauses quantify over countable ordinal indices and hence are first order over HC. Every descending ω-sequence of indices belongs to HC, so well-foundedness is first order as well. Quotient by the coded equality relation and take the transitive collapse.

Require c_0 to be empty and $c_{n+1} = c_n \cup \{c_n\}$ for $n < \omega$. Thus all natural-number constants have their actual values before the transfinite induction

begins. For each $0 < \alpha < \omega_1$, require that c_α be an ordinal, that $F_0(c_\alpha)$ have domain ω and range c_α, and that H_0 give the indicated inverse values. Require, for $n < \omega$,

$$F_0(c_\alpha)(c_n) = c_{f_{0,\alpha}(n)}.$$

The canonical surjections here are hereditarily countable, so these are first-order tests over HC. External induction on α identifies c_α with α. Require all these constants to belong to b_1. Since the collapse satisfies $b_1 = \omega_1$, it follows that b_1 is the true ω_1: it cannot be smaller because it contains every countable ordinal; if it were larger, the model would supply a surjection from ω onto the actual ω_1.

The clauses $z_i \subseteq b_1$ and $c_\alpha \in z_i \leftrightarrow \alpha \in X_i$ bind the input classes. The analogous clauses on the natural numbers bind the real parameters. Conversely, enumerate the universe of an actual model by the generators, interpret the ordinal constants by their actual values, and choose its Skolem functions. On ordinal constants choose the indicated canonical surjections. This gives the direct code. □

We now define the higher predicates simultaneously. All finitely many class witnesses used in one definition are packed into one class by the fixed pairing function. A variable E denotes an ordinal code, and $|E|_r$ denotes its actual order type, only in the metatheoretic verification.

Lemma 7.7. *For $r \geq 2$ there are predicates*

$$O_r, C_r^\varepsilon \in \Pi^2_{r-2} \quad (\varepsilon \in \{+,-\}), \qquad V_r \in \Pi^2_{r-1},$$

with the following properties.

(a) $O_r(E)$ means that $Z_{r-1}(E)$ codes a wellorder on a subset of κ_{r-1}, and $\{|E|_r : O_r(E)\} = \kappa_r$.

(b) If $O_r(E)$ and $a = |E|_r$, then

$$a \in Z_r(X) \iff \exists H\, C_r^+(X, E; H),$$
$$a \notin Z_r(X) \iff \exists H\, C_r^-(X, E; H).$$

Thus the signs are sound and exactly one has a witness.

(c) $V_r(Y; \vec{X}, \vec{t})$ means that $Z_r(Y)$ codes a well-founded extensional $N \models \mathsf{T}_r$ with

$$b_j^N = \kappa_j \ (j \leq r), \qquad z_\ell^N = Z_r(X_\ell), \qquad t_\ell^N = t_\ell.$$

Finitely many additional decoded sets may be bound likewise. Every such actual model code in the intermediate extensions of (C2) admits a representation Y satisfying V_r.

Proof. Induct on r, defining O_r, then $C_r^{\varepsilon}, W_r^{\theta}$, then V_r. Define $O_r(E)$ by the wellorder laws for $Z_{r-1}(E)$. For $r = 2$ these are first order. For $r > 2$, check the finite order laws universally through the lower signed witnesses and forbid one class packing a descending ω-sequence of ordinal codes and its positive entry witnesses. Thus $O_r \in \Pi^2_{r-2}$. Every $a < \kappa_r$ has a constructible wellorder code on a subset of κ_{r-1}; (C2) supplies E when $r > 2$, and $r = 2$ is direct.

A displayed H packs a V_{r-1}-code for a transitive model N, distinguished decoded inputs, a_i, full-relation isomorphisms π_i, and all positive witnesses for the finitely many required satisfaction entries and the particular test. Semantically,

$$\begin{gathered} \rho = \kappa_{r-1}, \qquad \beta = (\rho^+)^N, \\ z_i = Z_{r-1}(X_i) \in N, \qquad e_i = Z_{r-1}(E_i) \in N, \\ a_i < \beta, \qquad \pi_i : e_i \cong a_i. \end{gathered}$$

The V_{r-1} binding clauses fix the input copies. Every lower satisfaction-entry witness is already part of H; correctness introduces no existential quantifier for it. The model, matching, and entry tests are therefore Π^2_{r-2}. At $r = 2$, all model and entry codes are direct and these tests are first order.

For $C_r^{\varepsilon}(X, E; H)$, put $z = Z_{r-1}(X)$ and $a = |E|_r$, and test

$$\sup(z \cap A_a^{r-1}) < \rho \quad (\varepsilon = +), \qquad \sup(z \cap A_a^{r-1}) = \rho \quad (\varepsilon = -).$$

Totality and functionality of the canonical family over L^N_β, $a < \beta$, and upward Σ_1 absoluteness identify its value with the actual A_a^{r-1}. The tests are absolute, proving soundness. The same construction gives $W_r^{\theta} \in \Pi^2_{r-2}$ for equality and comparison of ordinals, bounds below true κ_j $(j < r)$, and values of the fixed canonical coding functions and $f_{j,a}$. These are fixed absolute tests, not arbitrary N-truth.

For exhaustiveness, use (C1) to localize the finitely many input classes and work over the appropriate upper extension with its finitely many retained lower levels. Let $\mathcal{G}$ include all full canonical graphs, including pairing and term constructors. Choose

$$\begin{gathered} M \prec H_\vartheta, \quad |M| = \rho, \quad \rho \cup \bigcup_i (a_i + 1) \subseteq M, \\ \rho, \vec{z}, \vec{e}, \vec{a}, \vec{\pi}, L_{\kappa_r}, \mathrm{Sat}(L_{\kappa_r}), \mathcal{G} \in M. \end{gathered}$$

For its collapse $\pi : M \to N$, condensation gives

$$\pi(\rho) = \rho, \qquad \beta = \pi(\kappa_r) = (\rho^+)^N > \max_i a_i, \qquad \pi(L_{\kappa_r}) = L_\beta,$$

and transfers totality of the displayed graphs. The model and its code belong to this intermediate extension. For $r > 2$, apply (C2) and rank-$(r-1)$

representation to its code; the lower-rank predicates supply the displayed entry witnesses. At $r = 2$, model and entry codes are direct.

Finally define V_r by universal signed readings of equality, membership and satisfaction. Require equality, congruence, the Skolemized Tarski clauses and T_r. For the satisfaction column S, use

$$\begin{aligned} S(\exists v\, \varphi(v, \vec{u})) &\Rightarrow S(\varphi(f_\varphi(\vec{u}), \vec{u})), \\ S(\varphi(w, \vec{u})) &\Rightarrow S(\exists v\, \varphi(v, \vec{u})) \quad \text{(all formal terms } w\text{)}. \end{aligned}$$

Check term constructors and column indices by W_r^θ. Each finite clause A is

$$\forall \vec{E}\, \forall \vec{H}\, [\mathrm{Corr}_r(\vec{E}, \vec{H}) \Rightarrow A], \qquad \mathrm{Corr}_r \in \Pi^2_{r-2},$$

where Corr_r conjoins correctness of ordinal codes and signed witnesses. Thus the clause is Π^2_{r-1}. A descending ω-sequence and its positive membership witnesses pack into one class, making ill-foundedness Σ^2_{r-1} and its negation Π^2_{r-1}.

Require, through these signed clauses,

$$c_0 = \varnothing, \qquad c_{n+1} = c_n \cup \{c_n\}\ (n < \omega), \qquad c_\alpha \text{ ordinal}.$$

Thus the finite constants are actual naturals. For $j < r$, $0 < \alpha < \kappa_{j+1}$ and $i < \kappa_j$, require

$$\begin{gathered} c_\alpha \in b_{j+1}, \qquad F_j(c_\alpha) : b_j \twoheadrightarrow c_\alpha, \\ u \in c_\alpha \Rightarrow \big[H_j(c_\alpha, u) \in b_j \ \wedge\ F_j(c_\alpha)(H_j(c_\alpha, u)) = u\big], \\ F_j(c_\alpha)(c_i) = c_\gamma \quad (\gamma = f_{j,\alpha}(i)). \end{gathered}$$

The last equation is required for every correct canonical-value witness. The bounds on α use W_r^θ for $j+1 < r$ and all valid ordinal codes for $j+1 = r$; the same convention checks $i < \kappa_j$. Universally quantify the displayed witnesses and terms u. The inverse term represents an alleged member of c_α; no new ordinal code is existentially quantified.

After collapse, $b_0 = \omega$. If $b_j = \kappa_j$, external induction on $0 < \alpha < \kappa_{j+1}$ gives

$$c_\alpha = \{c_{f_{j,\alpha}(i)} : i < \kappa_j\} = \{f_{j,\alpha}(i) : i < \kappa_j\} = \alpha.$$

Thus $b_{j+1} \geq \kappa_{j+1}$. Strict inequality, together with $N \models b_{j+1} = (b_j)^+$, would give an actual surjection $\kappa_j \twoheadrightarrow \kappa_{j+1}$ in N. Hence $b_j = \kappa_j$ for all $j \leq r$.

Bind X_i by $z_i \subseteq b_r$ and, universally over valid ordinal codes and correct pairs of signed witnesses, agreement of

$$|E|_r \in Z_r(X_i) \quad \Longleftrightarrow \quad q_{\mathrm{mem}}(c_{|E|_r}, z_i) \in Z_r(Y).$$

Check its index by W_r^θ. Exhaustiveness and the true b_r give $z_i = Z_r(X_i)$. The analogous clauses on natural-number constants bind the reals. These binding clauses are Π^2_{r-1}.

Conversely, enumerate an actual suitable model by its generators, use the actual ordinal constants, and choose Skolem functions, canonical surjections and inverse values. Its complete equality, membership and satisfaction code lies in $\mathcal{P}(\kappa_r)$. If this code lies in the specified intermediate extension, (C2) gives Y decoding to it; signed soundness and exhaustiveness give $V_r(Y;\vec{X},\vec{t})$.

This defines and verifies, successively, O_r, then $C_r^{\varepsilon}, W_r^{\theta}$, then V_r. The operations are recursive on finite formula codes. Decoded sets and actual cardinals above describe semantics; syntactically use the signed tests and distinguished constants. No true-cardinal or truth predicate is an extra parameter. Countably many classes for sequences or axiom schemes pack into one using $\kappa_1 \times \omega = \kappa_1$. □

7.4 The local club formula

For each $r \geq 1$ fix the following single first-order formula $\chi_r(\rho, \beta, z, t)$. It asserts that there are sets P, C such that

$$P = \{\eta < \beta : z \cap A_\eta^r \text{ is bounded in } \rho\}, \qquad C = \mathrm{Even}(P),$$

where the almost-disjoint graph is evaluated over the displayed L_β, and that $\psi_{\mathrm{src}}(\rho, \beta, C, t)$ holds. The formula numbers and the distinguished argument terms are fixed. Write q_r for the canonical index of this satisfaction entry. Thus the value of q_r is fixed by the chosen column and term conventions; it is not an index chosen by a proposed model code.

At level one, let $D_1(Y,t)$ be the direct assertion that the satisfaction entry with index q_1 is positive. For $r \geq 2$ put

$$D_r(Y,t) \quad \Longleftrightarrow \quad \exists E, H, K \left[W_r^{\mathrm{index}=q_r}(E;K) \wedge C_r^{+}(Y,E;H) \right].$$

The three class variables can be packed into one. Hence D_1 is first order and $D_r \in \Sigma^2_{r-1}$ for $r \geq 2$. For a valid $V_r(Y;X,t)$ code with collapse N,

$$D_r(Y,t) \quad \Longleftrightarrow \quad N \models \chi_r(\rho, \beta, z, t).$$

Moreover $\rho = \kappa_r$, $z = Z_r(X)$, and totality of the canonical graph identifies the displayed P in N with $Z_{r+1}(X) \cap \beta$. This is the binding needed in both directions of exact decoding.

For $r \geq 1$, put

$$\mathrm{Code}_{r+1}(t) \quad \Longleftrightarrow \quad \exists X \, \forall Y \, [V_r(Y;X,t) \longrightarrow D_r(Y,t)].$$

For $r = 1$, $V_1(Y;X,t) \to D_1(Y,t)$ is first order; for $r \geq 2$, V_r is Π^2_{r-1} and D_r is Σ^2_{r-1}. Thus $\mathrm{Code}_{r+1} \in \Sigma^2_{r+1}$, uniformly in r.

Proposition 7.8. *In the final model, for every $n \geq 1$ and $t \in I$,*

$$\mathrm{Code}_n(t) \quad \Longleftrightarrow \quad t \in T_n.$$

Proof. For $n = 1$, an added avoiding club proves the forward coding direction, and stationarity of every omitted S^1_t gives the converse.

Let $n = r+1 \geq 2$ and $t \in T_n$. Choose the avoiding club C and its padded target $P = P^n_t(C) \subseteq \kappa_n$. Successive coding gives X with $Z_n(X) = P$. Let $V_r(Y; X, t)$ hold, and let N be its transitive collapse. The model has true $\rho = \kappa_r$, internal successor $\beta < \kappa_n$, and contains $z = Z_r(X)$. Its local almost-disjoint decoding is exactly $P \cap \beta$ and belongs to N by Separation. The padding lemma therefore gives $N \models \psi_{\mathrm{src}}(\rho, \beta, C \cap \beta, t)$. Thus $N \models \chi_r$, and $D_r(Y, t)$ follows. Since this holds for every valid Y, X witnesses $\mathrm{Code}_n(t)$.

Conversely let $t \notin T_n$, and suppose that X witnesses $\mathrm{Code}_n(t)$. By (C1), after all coordinates of levels at least r have been forced, X belongs to the extension by a retained complete restriction on the finitely many lower levels. Work in this intermediate model. For every $\delta < \kappa_n$, choose an elementary submodel of a sufficiently large H_θ, of size κ_r, containing κ_r pointwise, $\delta + 1$ pointwise, the distinguished input $z = Z_r(X)$, the real t, L_{κ_n}, its satisfaction relation, and the full canonical graphs, including the pairing and term constructors. Its transitive collapse N_δ has true κ_r and internal successor $\beta_\delta > \delta$, and satisfies T_r.

The full code of N_δ is a subset of κ_r in this intermediate extension. For $r \geq 2$, include all coordinates of level $r - 1$ in the upper extension; the remaining restriction is on levels below $r - 1$ and still satisfies the cardinal bounds in (C2). That assertion and the representation direction of the coding lemma give Y_δ with $V_r(Y_\delta; X, t)$. For $r = 1$ use its direct code. The defining property of X gives $D_r(Y_\delta, t)$.

Put $P = Z_n(X)$ in the ambient model. The verified satisfaction entry says that $\mathrm{Even}(P) \cap \beta_\delta$ is club in β_δ and avoids $S^n_t \cap \beta_\delta$. These are initial segments of the same fixed set, for cofinally many $\beta_\delta < \kappa_n$. Consequently $\mathrm{Even}(P)$ is unbounded and closed in κ_n and disjoint from S^n_t: unboundedness and avoidance follow by taking a prefix above any specified ordinal, and closure follows by taking such a prefix above any limit point. This contradicts (C3). □

8 Comparing the club-forcing extensions

Put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$ and write $M_{\vec{T}} = B_{\vec{T}}[G_{D_{\vec{T}}}]$. Throughout this section use the same constructible stationary sets and the same ground partial orders in all products. By Proposition 6.7, for each Π^2_n formula $A(a)$ there is a $\Pi^2_1(H_{\kappa_n})$ formula $\Phi_A(a)$, independent of $\vec{T}$, such that

$$M_{\vec{T}} \models A(a) \quad \Longleftrightarrow \quad B_{\vec{T}} \models \Phi_A(a).$$

The corresponding assertion holds for Σ^2_n formulas. In particular, this equivalence holds for every generic filter for $D_{\vec{T}}$. No comparison between $D_{\vec{R}}$ and $D_{\vec{T}}$ will be needed.

Whenever $\vec{R} \leq \vec{T}$, choose the club generics by restriction of a single $C_{\vec{T}}$-generic filter. This gives $B_{\vec{R}} \subseteq B_{\vec{T}}$. Results about the forcing decisions below do not depend on this choice of compatible generics.

Lemma 8.1. *Suppose that $\vec{R} \leq \vec{T}$ differ only at coordinate n. For compatible club generics,*

$$H_{\kappa_n}^{B_{\vec{R}}} = H_{\kappa_n}^{B_{\vec{T}}}.$$

Proof. Put $\kappa = \kappa_n$ and factor the club product in the following order: the common factors above n, the common factor $C_n(R_n)$, the omitted factor $C_n(T_n \setminus R_n)$, and finally

$$P = \prod_{k<n} C_k(T_k).$$

Here P is trivial if $n = 1$, and otherwise has cardinality $\kappa_{n-1} < \kappa$. The equality for the row-n product is a literal product factorization; for $n = 1$, the union of two countable supports is countable.

Over the common upper extension, both $C_n(R_n)$ and $C_n(T_n) = C_n(R_n) \times C_n(T_n \setminus R_n)$ are $< \kappa$-distributive by Lemma 3.3 and its row factorization proof. Consequently, after the common row-n generic has been fixed, the omitted factor is $< \kappa$-distributive. Indeed, neither the intermediate nor the full row extension adds ordinal sequences of length $< \kappa$ over the common upper model. By Lemma 2.4, the omitted factor remains $< \kappa$-distributive after the same P-generic filter is added. Thus the two final club-forcing extensions have the same ordinal sequences of length $< \kappa$.

Both preserve κ. A set of hereditary size $< \kappa$ can be coded by a well-founded extensional pointed relation on an ordinal $\gamma < \kappa$. Such a relation has an ordinal-sequence code of length $< \kappa$. The equality of these sequence collections therefore gives the displayed equality of H_κ. □

Proposition 8.2. *Suppose that $\vec{R} \leq \vec{T}$ differ only at n. If $A(a)$ is Π^2_n and $F_{\vec{T}} \Vdash A(a)$, then $F_{\vec{R}} \Vdash A(a)$.*

Proof. Work first with compatible club generics. The previous lemma gives a common first-order domain H_{κ_n}. Every class over this domain in $B_{\vec{R}}$ is also a class in $B_{\vec{T}}$, and the first-order formula following the class quantifier in Φ_A has the same interpretation on each old class. Hence

$$B_{\vec{T}} \models \Phi_A(a) \quad \Longrightarrow \quad B_{\vec{R}} \models \Phi_A(a).$$

Proposition 6.7 gives the asserted implication for the corresponding extensions by D.

For the forcing assertion, start with an arbitrary $C_{\vec{R}}$-generic filter and extend it to a $C_{\vec{T}}$-generic filter by the omitted product. The hypothesis and Proposition 6.7 give Φ_A in the larger club extension, and the preceding implication gives Φ_A in the original one. Since that club generic was arbitrary, Proposition 6.7 implies $F_{\vec{R}} \Vdash A(a)$. □

Proposition 8.3. *Suppose that $\vec{R}, \vec{T}$ agree through coordinate n. For every Σ^2_n or Π^2_n formula $A(a)$, the club-forcing extensions with compatible common generics give the same truth value to $\Phi_A(a)$. Consequently $F_{\vec{R}}$ and $F_{\vec{T}}$ force the same instances of such formulas.*

Proof. First suppose that $\vec{R} \leq \vec{T}$ and put $\kappa = \kappa_n$. Force with the common factors above n, then with the common factor $C_n(T_n)$, and call this extension Z. Put

$$R = \prod_{m>n} C_m(T_m \setminus R_m), \qquad P = \prod_{m<n} C_m(T_m).$$

The two models to be compared are $Z[G_P]$ and $Z[G_R][G_P]$, where (G_P, G_R) is product generic over Z.

The ground partial order R is $< \kappa$-closed: its factor at $m > n$ is $< \kappa_{m-1}$-closed and the product uses full support over m. The common upper product is also $< \kappa$-closed, and the common row-n forcing is $< \kappa$-distributive over its extension. They therefore add no ordinal sequences of length $< \kappa$. Every sequence of fewer than κ conditions of the old R still belongs to the ground model, by coding its terms with a fixed ground enumeration. Its ground lower bound is still a lower bound. Thus R is $< \kappa$-closed in Z as well.

The model Z satisfies $\kappa^{<\kappa} = \kappa$, by Lemma 3.3 applied to the sequence with empty supports below n. Moreover, $|P| < \kappa$, so P is κ-cc and has density at most κ. For each fixed $p \in P$, relative existential second-order absoluteness and its universal dual give

$$Z \models p \Vdash_P \Phi_A(a) \quad \iff \quad Z[G_R] \models p \Vdash_P \Phi_A(a).$$

For a Σ^2_1 formula Φ_A use the existential assertion directly; for a Π^2_1 formula Φ_A use its universal dual. The real parameters and their constant names are the same in both models.

Choose a condition in G_P deciding $\Phi_A(a)$ over Z. The same condition makes the same decision over $Z[G_R]$. The forcing theorem therefore gives equivalence of $\Phi_A(a)$ in the two club extensions. Proposition 6.7 gives the corresponding equivalence for $A(a)$ after D.

Any generic for the smaller full club product can be extended to a generic for the larger product. Reordering that product produces the generics just used. Thus an instance forced on one side is forced on the other side. For incomparable sequences compare both with their coordinatewise intersection; their first n supports remain the same. This also covers changes at infinitely many higher coordinates. □

Proposition 8.4. *For every $\vec{T} \in L$, the forcing $F_{\vec{T}}$ decides each finite second-order formula with real parameters. It also preserves cardinals and adds no reals. Thus it satisfies (A1).*

Proof. Every finite second-order formula can be put in Σ^2_n or Π^2_n form for some finite n. By Proposition 6.7, its truth in the extension by $D_{\vec{T}}$ is determined in $B_{\vec{T}}$ by a fixed formula whose only nonlogical parameters are constructible reals. Cone homogeneity of $C_{\vec{T}}$ implies that its largest condition decides this formula: opposite decisions below two conditions would remain opposite below isomorphic stronger cones, a contradiction. Proposition 6.7 then gives a decision by the largest condition of $F_{\vec{T}}$.

The club product preserves cardinals and adds no reals. The coding forcing preserves cardinals by its structural theorem, and it is countably closed by the prescribed lower-bound construction. The two-step iteration therefore preserves cardinals and adds no reals. □

Remark 8.5. The transformation in Proposition 6.7 is recursive in the finite formula code, so all the decisions used here are uniformly definable in L. Truth of the resulting formula is interpreted in the set-sized two-sorted structure $(H_{\kappa_n}, \mathcal{P}(H_{\kappa_n}))$ of the relevant club extension. The ordinary forcing relation for these structures is uniformly definable. Hence this uniformity uses no truth predicate for the universe.

8.1 Verification of the four hypotheses

Theorem 8.6. *For $\vec{T} \in L$ put $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$. The uniformly defined family $\langle F_{\vec{T}} \rangle$ satisfies (A1)–(A4) of Theorem 2.2. Its forcing decisions are uniformly definable in L from the finite formula codes and the original real parameters.*

Proof. Proposition 8.4 gives (A1), including cardinal preservation and the absence of new reals. The uniform parameter-free coding predicates of Proposition 7.8 give (A2). Propositions 8.2 and 8.3 give (A3) and (A4), respectively; the latter allows changes at infinitely many higher coordinates. Remark 8.5 gives uniform definability of the forcing decisions. All these results precede the deletion recursion. □

9 The deletion recursion and the uniformizing functions

The following recursion and verification prove Theorem 2.2 from (A1)–(A4), independently of the particular forcing construction. Theorem 8.6 supplies the hypotheses for $C_{\vec{T}} * \dot{D}_{\vec{T}}$.

9.1 The recursion in L

Fix a recursive enumeration $\langle A_{n,e} : e < \omega \rangle$ of the Π^2_n formulas $A_{n,e}(p, x, y)$. Work in L, and put

$$\nu_{n,e}(\vec{T}, p, x, y) = 1 \quad \Longleftrightarrow \quad 1_{F_{\vec{T}}} \Vdash A_{n,e}(p, x, y).$$

By (A1), every instance has value zero or one. Call the section indexed by (e, p, x) *nonempty at* $\vec{T}$ if $\nu_{n,e}(\vec{T}, p, x, y) = 1$ for some $y \in \mathbb{R}^L$. This agrees with nonemptiness in any $F_{\vec{T}}$ extension. Such an extension has no new reals, so any witness is a ground-model real. Since its instance is decided, a condition forcing that instance implies that the largest condition forces it. No uniform satisfaction predicate for the universe is used.

Put $\vec{T}^0 = \langle I : n \geq 1 \rangle$. For $n \geq 1$, keep all coordinates other than n fixed at their values in $\vec{T}^{n-1}$. At this point $T_n^{n-1} = I$. Set $\vec{T}^{n,0} = \vec{T}^{n-1}$ and begin with an empty partial function on the level-n sections. At stage α:

(i) if some section is nonempty at $\vec{T}^{n,\alpha}$ and has not yet been assigned a value, take the $<_L$-least such (e, p, x) and the $<_L$-least real y satisfying $\nu_{n,e}(\vec{T}^{n,\alpha}, p, x, y) = 1$. Assign y to (e, p, x) and remove only $\ell(e, p, x, y)$ from the nth support;

(ii) at a limit stage intersect the nth supports and take the union of the partial functions;

(iii) terminate the recursion if there is no unassigned nonempty section.

Write $\vec{T}^n$ for the sequence at termination and a_n for the resulting partial function. Finally put

$$T_n^* = T_n^n, \qquad \vec{T}_* = \langle T_n^* : n \geq 1 \rangle.$$

Lemma 9.1. *For every n, the recursion terminates before κ_2. It is definable in L, and $\vec{T}_* \in L$. Moreover, if $a_n(e, p, x) = y$, then*

$$\nu_{n,e}(\vec{T}^n, p, x, y) = 1.$$

Every section nonempty at the final support is assigned a value.

Proof. The least section and its least value exist by the definition of nonemptiness and the constructible wellorder. Every nontrivial successor step assigns a new section. There are only κ_1 sections in L, so this cannot occur at every stage below κ_2.

Once a value y has been assigned to (e, p, x), (A3) ensures that $A_{n,e}(p, x, y)$ is forced at every later support in this recursion. At a limit stage apply (A3) directly to the support where y was chosen and the intersection support. At termination there is no unassigned nonempty section.

Uniform definability of the forcing decisions makes this a definable set recursion in L. Replacement over $n \in \omega$ gives the sequence $\langle \vec{T}^n : n \geq 1 \rangle$ and the final sequence $\vec{T}_*$. □

9.2 Verification

Proof of Theorem 2.2. Fix n. The sequences $\vec{T}^n$ and $\vec{T}_*$ agree through coordinate n. By (A4),

$$\nu_{n,e}(\vec{T}^n, p, x, y) = \nu_{n,e}(\vec{T}_*, p, x, y)$$

for every real triple. Thus they also give the same nonempty sections, since the reals are unchanged. Here (A4) applies to the entire sequence of higher supports at once; no passage through successive preservation statements is needed.

In an $F_{\vec{T}_*}$ extension define

$$U_{n,e}(p, x, y) \quad \iff \quad A_{n,e}(p, x, y) \ \wedge \ \neg\, \mathrm{Code}_n(\ell(e, p, x, y)).$$

This is Π^2_n. By (A2) and injectivity of ℓ,

$$\neg\, \mathrm{Code}_n(\ell(e, p, x, y)) \quad \iff \quad a_n(e, p, x) = y.$$

Indeed, the nth support initially contains every code, and exactly the code of the chosen value is removed at each assigned section. The other recursions leave the nth support unchanged. The lemma and (A4) ensure that each chosen pair still belongs to $A_{n,e}$. Every nonempty final section is assigned a value, and every unassigned final section is empty. Consequently $U_{n,e}$ is a subset of $A_{n,e}$ and has exactly one value on every nonempty section.

This treats every n, e and every real parameter, because all final reals are constructible. The defining formula uses only the original parameter p. The sequence $\vec{T}_*$ and the partial function a_n are used in constructing the forcing and verifying the formula, but neither is a parameter of $U_{n,e}$. □

Proof of Theorem 1.1. Apply Theorem 2.2 to the family $F_{\vec{T}} = C_{\vec{T}} * \dot{D}_{\vec{T}}$, whose hypotheses are verified in Theorem 8.6. The final sequence $\vec{T}_*$ belongs to L, and $F_{\vec{T}_*}$ is the required cardinal-preserving set forcing which adds no reals. □

10 Projective determinacy

We first carry out the construction with a fixed predicate $A \subseteq \omega_1$. We then recover the predicate needed for $L(\mathbb{R})^{M_\omega}$ by parameter-free Δ^2_1 definitions.

10.1 The construction over a relative constructible ground model

In this subsection fix

$$\mathcal{W} = L[A] \models \mathsf{ZFC}, \qquad A \subseteq \omega_1^{\mathcal{W}}, \qquad \kappa_n = \omega_n^{\mathcal{W}}.$$

All ground-model objects and cardinals are now computed in $\mathcal{W}$. Write $\Sigma^2_n[A]$ and $\Pi^2_n[A]$ for the pointclasses over $(\mathrm{HC}, \in, A)$, with class quantifiers ranging over all subsets of HC in the ambient extension. The predicate A always denotes the same ground-model set. Additional parameters are reals.

Lemma 10.1. *The stationary families, almost disjoint families and padding construction of Section 3 have relative versions in $\mathcal{W}$. At level one the source graph is first order over $(\mathrm{HC}, \in, A)$ and is unchanged in extensions preserving* HC*. At level $n \geq 2$, put $\rho = \kappa_{n-1}$ and $\nu = \kappa_n$. The canonical graphs are $\Sigma_1(A)$ over $L_\nu[A]$, and Lemma 3.1 holds with $L_\beta[A]$ and $L_{\nu \cdot 2+1}[A, C]$ in place of the corresponding constructible levels, provided the local transitive models contain the actual A.*

Proof. Relative condensation gives GCH in $\mathcal{W}$ and

$$H^{\mathcal{W}}_{\kappa_n} = L_{\kappa_n}[A] \qquad (n \geq 1).$$

For $n \geq 2$, use hulls of size ρ containing ρ pointwise and A as an element. Their collapses fix $A \subseteq \omega_1$ and place every member of $H^{\mathcal{W}}_\nu$ in $L_\nu[A]$. For $x \in \mathrm{HC}^{\mathcal{W}}$, take a countable hull containing A, ω_1 as elements and the transitive closure of x pointwise. Its collapse places x in $L_\beta[A \cap \delta]$ for countable β, δ. The countable relative construction belongs to a further countable stage of $L[A]$, proving the equality also at $n = 1$. The usual counting of these relative levels gives CH; above ω the hulls fixing A give $2^\rho = \rho^+$.

Use the good Σ_1 order $<_{L[A]}$ throughout. At a countable limit δ, define d_δ by the least pair (u, c) in $L[A \cap \delta]$ such that

$$u \subseteq \delta, \quad c \subseteq \delta \text{ is club}, \quad \forall \xi \in c \; (u \cap \xi \neq d_\xi),$$

and put $d_\delta = u$; use the empty set if no pair exists and at nonlimits. Every local comparison and recursion history has a countable construction certificate, so the graph is first order over $(\mathrm{HC}, \in, A)$ and is unchanged when this structure is unchanged.

If (X, C) were the $<_{L[A]}$-least counterexample to diamond, collapse a countable elementary hull of a sufficiently large limit level $L_\theta[A]$ containing $A, \omega_1, \vec{d}, X, C$. For δ its intersection with ω_1, the collapsed predicate is $A \cap \delta$, and the history collapses to $\vec{d} \restriction \delta$. The collapsed counterexample is the least local eligible pair: all its predecessors in the stage-first order are in the collapsed level, and eligibility is absolute there. Hence $d_\delta = X \cap \delta$, contradicting $\delta \in C$. Its real traces give the level-one stationary family and reserve.

At $\nu = \rho^+ \geq \omega_2$, run the diamond recursion on $\nu \cap \mathrm{cof}(\rho)$ using $<_{L[A]}$. The proof of diamond now uses hulls containing A as an element and ρ pointwise, so their collapses keep the same predicate A. The leastness witnesses and histories belong to $L_\nu[A]$. Thus the source graph has its relative $\Sigma_1(A)$ definition. Define its stationary real traces and reserve as in Section 3. The

enumeration of $\mathcal{P}(\rho)^{\mathcal{W}}$ and the almost disjoint family are obtained using the same relative order and have the corresponding $\Sigma_1(A)$ graphs.

For an avoiding club $C \subseteq \nu \setminus S^n_t$, use

$$K = L_{\nu\cdot 2+1}[A, C].$$

Choose its elementary chain in $L[A, C]$ with A, ρ, ν, C, t as elements of every member and ρ pointwise included. Keep the same even-part coding and odd-block padding, using $<_{L[A]}$ for the block wellorders. If $\beta = (\rho^+)^N < \nu$ and the padded target restricted to β belongs to a suitable transitive N containing A, the odd-block argument gives $\beta \in E$. Relative condensation then gives

$$\bar{K} = L_{\beta\cdot 2+1}[A, C \cap \beta] \subseteq N.$$

Collection and the relative-hierarchy recursion in the suitable theory put this level in N. The relative source graphs have their actual values in $L_\beta[A]$ by upward $\Sigma_1(A)$ absoluteness and functionality. Elementarity therefore gives the local avoiding-club conclusion. Conversely, the test hulls fix A and give the actual source and almost disjoint families; the cofinal collection of local even parts reconstructs an avoiding club. This is the proof of Lemma 3.1 with the indicated relative levels. □

Lemma 10.2. *The model predicates of Section 7 can require that every coded model contains the actual A, with the same complexity bounds:*

$$V_1^A \text{ is first order}, \qquad O_r^A, C_r^{A,\pm} \in \Pi^2_{r-2}[A], \qquad V_r^A \in \Pi^2_{r-1}[A] \quad (r \geq 2).$$

Proof. Add a distinguished object a to each suitable theory, replace L_β by $L_\beta[a]$, and replace its constructible-cardinal assertions by the corresponding $L[a]$ assertions. The canonical surjections and other graphs are defined using $<_{L[a]}$.

For a direct code Y with equality $=_Y$ and membership E_Y, the ordinal constants may be identified by the first-order clauses

$$E_Y(u, c_\xi) \iff \exists \eta < \xi \ (u =_Y c_\eta) \qquad (\xi < \omega_1).$$

After well-founded collapse, induction gives $c_\xi = \xi$. Requiring all these constants to belong to b_1, together with $N \models b_1 = \omega_1$, gives the true $b_1 = \omega_1$. A larger value would give a surjection from ω onto the actual ω_1. Now require

$$N \models a \subseteq b_1, \qquad E_Y(c_\xi, a) \iff A(\xi) \quad (\xi < \omega_1). \tag{P1}$$

These are first-order conditions over $(\mathrm{HC}, \in, A)$ and imply $a^N = A$. This gives the relative version of Lemma 7.6.

Proceed through Lemma 7.7 by induction. The lower model predicates already bind the actual A, so their relative canonical graphs have the actual values. At rank r, read (P1) through the same universal signed membership

and satisfaction clauses used to bind the other inputs. Here ξ is a countable ordinal and $A(\xi)$ is atomic. No additional class block is introduced. The original rank calculation gives the displayed bounds. In the representation direction, include A in each hull containing κ_r pointwise; its collapse fixes A and satisfies (P1). □

Theorem 10.3. *There is a cardinal-preserving forcing $\mathbb{P}_A \in \mathcal{W}$ adding no reals and forcing $\Pi^2_n[A]$-UP for every finite $n \geq 1$. Each uniformizer uses only A and the original real parameters.*

Proof. Let I consist of the real labels from $\mathcal{W}$ and carry out the construction of Sections 4 and 5 inside $\mathcal{W}$. Use the families from Lemma 10.1 and $<_{L[A]}$ for all fixed choices. Conditions, support rules, dependencies, term forcings and complete restrictions have the same definitions. Their bounds use GCH, closure, linking and the stationary reserves, all available in $\mathcal{W}$. Thus the structural theorem 5.10 and its small-quotient and localization lemmas hold with this ground model.

Write

$$F^A_{\vec{T}} = C^A_{\vec{T}} * \dot{D}^A_{\vec{T}}, \qquad B^A_{\vec{T}} = \mathcal{W}[G_{C^A_{\vec{T}}}].$$

The proof of Proposition 6.7 admits fixed ground predicates: Fact 2.5 and Lemma 2.6 already allow them, and the name-table translation carries the fixed name for A. Hence each $\Pi^2_n[A]$ formula φ has a $\Pi^2_1(H_{\kappa_n}, A)$ translation Φ^A_φ, independent of $\vec{T}$, satisfying

$$B^A_{\vec{T}}[G_{D^A_{\vec{T}}}] \models \varphi(p, x, y) \iff B^A_{\vec{T}} \models \Phi^A_\varphi(p, x, y).$$

Lemmas 10.1 and 10.2, followed by the proof of Proposition 7.8, give

$$\mathrm{Code}^A_n \in \Sigma^2_n[A], \qquad F^A_{\vec{T}} \Vdash (\mathrm{Code}^A_n(t) \iff t \in T_n). \tag{P2}$$

The cone isomorphisms of the club product fix $\check{A}$ and all ground reals. Proposition 6.7 therefore gives (A1), and (P2) gives (A2). For (A3), removing coordinates only at level n leaves the relevant H_{κ_n} unchanged, by the proof of Lemma 8.1. The predicate A has the same interpretation, so downward Π^2_1 absoluteness applies to the translated assertion. For (A4), the omitted higher product is $< \kappa_n$-closed, while the common lower product has size $< \kappa_n$. Apply Lemma 2.6 with $\check{A}$ among its predicate names, as in Proposition 8.3. This treats all higher coordinates together. The forcing decisions are uniformly definable in $\mathcal{W}$ by Remark 8.5.

The proof of Theorem 2.2 now runs in $\mathcal{W}$: replace the constructible labels by I and the least choices by $<_{L[A]}$-least choices. There are κ_1 sections, each level terminates before κ_2, and Replacement gives the terminal sequence $\vec{T}_* \in \mathcal{W}$. For $\mathbb{P}_A = F^A_{\vec{T}_*}$, the uniformizers are

$$U_{n,e}(p, x, y) \iff \varphi_{n,e}(p, x, y) \wedge \neg\, \mathrm{Code}^A_n(\ell(e, p, x, y)).$$

The verification in Section 9 uses only (A1)–(A4), so it gives the conclusion, including preservation of cardinals and the absence of new reals. □

10.2 Eliminating the fixed predicate

Lemma 10.4. *Suppose membership and nonmembership in $A \subseteq \omega_1$ have complementary Σ^2_1 definitions with a fixed real parameter p_0 (possibly absent). Then every $\Sigma^2_n[A]$ or $\Pi^2_n[A]$ formula, $n \geq 1$, has an equivalent ordinary formula at the same level using p_0. The assertion holds in every model where the two definitions are correct, also in the presence of additional free class variables.*

Proof. Write the definitions as $\exists X\, \theta_+(\xi, X, p_0)$ and $\exists X\, \theta_-(\xi, X, p_0)$, where θ_+ and θ_- are first order. Choice collects their witnesses for all $\xi < \omega_1$ into one class. Thus there is a first-order condition $D(B, Z, p_0)$ asserting $B \subseteq \mathrm{Ord} \cap \mathrm{HC}$ and

$$\forall \xi < \omega_1 \big[(\xi \in B \to \theta_+(\xi, Z_\xi, p_0)) \wedge (\xi \notin B \to \theta_-(\xi, Z_\xi, p_0))\big],$$

such that $\exists Z\, D(B, Z, p_0)$ holds exactly for $B = A$. For an existential-leading formula $\varphi(A)$ use

$$\exists B \exists Z\, [D(B, Z, p_0) \wedge \varphi(B)];$$

for a universal-leading formula use

$$\forall B \forall Z\, [D(B, Z, p_0) \to \varphi(B)].$$

The added class quantifiers merge with the first block. □

10.3 The ground model and its code

Assume the large-cardinal hypothesis of Theorem 1.2. Let $\mathcal{M} = M_\omega$, $\mathcal{R} = \mathbb{R}^{\mathcal{M}}$, and $\mathcal{W} = L(\mathcal{R})^{\mathcal{M}}$. Initially we use only $\mathcal{W} \models \mathsf{ZF}$. For the code below identify reals with subsets of ω, using the fixed recursive coding when the relation is on ω^ω. In first-order definitions over $\mathcal{W}$, the whole set $\mathcal{R}$ is named, as in [19, Section 7, printed p. 66].

Lemma 10.5. *There is $A \subseteq \kappa = \omega_1^{\mathcal{W}}$ such that*

$$\mathcal{W} = L[A] \models \mathsf{ZFC} + \mathsf{CH} + \mathsf{PD}.$$

Both membership and nonmembership in A, on countable ordinal inputs, are Σ_1 over $\mathcal{W}$ with $\mathcal{R}$ named, without additional parameters.

Proof. Use the natural construction order $\prec$ of $\mathcal{R}$. Steel states that this order is Δ_1 over $\mathcal{W}$ in the paragraph after Corollary 7.21 [19, printed p. 78]. Its graph is a set of $\mathcal{W}$ by Separation. Acceptability gives

$$\mathcal{P}(\omega)^{\mathcal{M}} \subseteq J^{\mathcal{M}}_{\omega_1^{\mathcal{M}}};$$

see [19, Definition 2.3 and the following paragraph]. The hierarchy is continuous, and each $J_\beta^{\mathcal{M}}$ for $\beta < \omega_1^{\mathcal{M}}$ is countable in $\mathcal{M}$, by its construction under countably many finitary rudimentary functions. Thus each proper initial segment of the stage-first order is countable in $\mathcal{M}$. Its countability witness is coded by a real, hence belongs to $\mathcal{W}$. The models have the same reals and consequently the same countable well-founded codes, HC and ω_1.

Every proper initial segment of $\prec$ is therefore countable in $\mathcal{W}$. In ZF its ordinal collapse has type at most $\omega_1^{\mathcal{W}}$, and Cantor's theorem makes the type uncountable. Thus the unique increasing enumeration

$$e : \kappa \longrightarrow \mathcal{R}$$

belongs to $\mathcal{W}$. No choice or CH assumption in $\mathcal{W}$ has been used. Put

$$A = \{\omega \cdot \xi + n : \xi < \kappa,\ n \in e(\xi)\}. \tag{P3}$$

Then $A \in \mathcal{W}$ and $L[A] \subseteq \mathcal{W}$. Conversely $L[A]$ recovers every row $e(\xi)$, hence e and its range $\mathcal{R}$ as sets. Minimality of $L(\mathcal{R})$ gives $\mathcal{W} = L[A]$. This proves choice in $\mathcal{W}$, and e gives CH. Every finite projective determinacy fragment holds in $\mathcal{M}$ by the projective determinacy theorem [15], using finitely many Woodin cardinals and a measurable below a later Woodin cardinal. The same reals give the same projective payoffs and strategy codes, so $\mathcal{W} \models \mathsf{PD}$.

For the complexity calculation, let $E(\xi, r)$ assert

$$r \in \mathcal{R} \ \wedge\ \exists f\,[f : (\xi, \in) \cong (\{s \in \mathcal{R} : s \prec r\}, \prec)].$$

The assertion that f is the displayed isomorphism uses only quantifiers bounded by $\mathcal{R}, \xi, f$ and the Δ_1 order. Collection gives closure under these bounded quantifiers; hence E is Σ_1 over $\mathcal{W}$ with $\mathcal{R}$ named. For $\xi < \kappa$ it says exactly $r = e(\xi)$. For $\alpha = \omega \cdot \xi + n < \kappa$,

$$\begin{aligned} \alpha \in A &\iff \exists r \in \mathcal{R}\,[E(\xi, r) \wedge n \in r], \\ \alpha \notin A &\iff \exists r \in \mathcal{R}\,[E(\xi, r) \wedge n \notin r]. \end{aligned} \tag{P4}$$

Ordinal division by ω is Δ_1-definable in ZF. Both formulas in (P4) are therefore Σ_1 and are complementary by totality and uniqueness of e. Substitute the fixed definitions of $\prec$; neither e, κ nor the order graph is an additional parameter. □

10.4 Recovery in the forcing extensions

Lemma 10.6. *Suppose $\mathcal{V} \models \mathsf{ZFC} + \mathsf{CH}$, put $\mathcal{R}_{\mathcal{V}} = \mathbb{R}^{\mathcal{V}}$, and assume*

$$\mathcal{L} = L(\mathbb{R})^{\mathcal{V}} \models \mathsf{ZFC} + \mathsf{CH}, \qquad \mathrm{HC}^{\mathcal{L}} = \mathrm{HC}^{\mathcal{V}}.$$

For every Σ_1 formula $\varphi(\mathcal{R}_{\mathcal{V}}, a)$ there is a uniformly obtained Σ^2_1 formula $\widehat{\varphi}(a)$ such that

$$\mathcal{L} \models \varphi(\mathbb{R}^{\mathcal{V}}, a) \iff \mathcal{V} \models \widehat{\varphi}(a) \qquad (a \in \mathrm{HC}^{\mathcal{V}}).$$

The translation has no predicate parameter for $\mathbb{R}^{\mathcal{V}}$.

Proof. Write φ as $\exists z\, \psi(z, \mathcal{R}_\mathcal{V}, a)$ with bounded ψ, and put $H = \mathrm{HC}^\mathcal{V}$ and $\tau = \omega_1^\mathcal{V} = \omega_1^\mathcal{L}$. If it holds in $\mathcal{L}$, choose a sufficiently large relative level $L_\theta(\mathcal{R}_\mathcal{V})$ containing a witness, H and the required construction histories. Inside $\mathcal{L}$ take an elementary hull of size τ containing $\mathcal{R}_\mathcal{V}, H, a$ as elements and every member of H pointwise. The transitive base $\mathrm{TC}(\{\mathcal{R}_\mathcal{V}\})$ is fixed pointwise by its collapse. Condensation above this base gives a level $L_\beta(\mathcal{R}_\mathcal{V})$ of size at most τ, $\beta < \tau^+$, containing the actual $\mathcal{R}_\mathcal{V}, H, a$ and a witness for ψ.

The fixed-base condensation assertion follows by collapsing the relative hierarchy together with its satisfaction relations. At a successor, elementarity supplies both the defining formula and parameters for every set present and the set defined by each formula and parameter tuple. The collapsed successor is therefore exactly Def of the preceding collapsed level. At limits the collapsed levels are unions. Induction identifies them with the actual relative hierarchy.

The level, its construction history and its satisfaction relations have hereditary size at most τ. One class $X \subseteq H$ can therefore code them and the marked witness. Require a well-founded extensional relation on a field $D \subseteq H$, a distinguished copy of the actual H, and a marked $\mathcal{R}_\mathcal{V}$ whose members are exactly the copies of the actual reals. For the copy of H, use indices $\langle 0, u\rangle$ for all $u \in H$ and require exactly their actual predecessor sets. Require the hierarchy to start with $\mathrm{TC}(\{\mathcal{R}_\mathcal{V}\})$, to have exact Def successors, unions at limits and correct Tarski satisfaction clauses with standard formula numbers. Finally require the marked final level to contain a and a witness for $\psi(z, \mathcal{R}_\mathcal{V}, a)$.

All these conditions are first order over H: objects, stage indices, parameters and formula numbers have indices in H. Well-foundedness is first order as well, since it is

$$\neg\exists f \in H\, [f : \omega \to D \ \wedge\ \forall n < \omega\ f(n+1)\ E_X\ f(n)],$$

and every countable sequence of members of H belongs to H. Pack all components into the one class X.

Conversely, the collapse of any accepted certificate has the actual base. The exact successor and limit clauses identify all its levels with the actual $L_\gamma(\mathcal{R}_\mathcal{V})$ by external induction. Its witness belongs to $\mathcal{L}$, and bounded absoluteness gives $\mathcal{L} \models \psi(z, \mathcal{R}_\mathcal{V}, a)$. Thus the assertion that such a certificate X exists is the required Σ^2_1 formula. □

Lemma 10.7. *For the A in (P3), there are parameter-free Σ^2_1 formulas $\sigma_+(\alpha)$ and $\sigma_-(\alpha)$ such that, in every set-forcing extension $\mathcal{W}'$ of $\mathcal{W}$ preserving* HC*,*

$$\begin{aligned} \mathcal{W}' \models \sigma_+(\alpha) &\iff \alpha \in A, \\ \mathcal{W}' \models \sigma_-(\alpha) &\iff \alpha \notin A \end{aligned} \qquad (\alpha < \omega_1^\mathcal{W}).$$

Proof. Translate the two formulas (P4) by Lemma 10.6. Every such $\mathcal{W}'$ has the same reals and ordinals as $\mathcal{W}$, so

$$L(\mathbb{R})^{\mathcal{W}'} = L(\mathbb{R})^{\mathcal{W}} = \mathcal{W}.$$

The ground enumeration e still witnesses CH in $\mathcal{W}'$. Apply the translation lemma in $\mathcal{W}'$ with this same inner model $\mathcal{W}$. The two first-order definitions still refer to the same order and enumeration, and consequently recover the original A. The soundness direction excludes false witnesses from new class codes. □

Proof of Theorem 1.2. Take $\mathcal{W} = L[A]$ from Lemma 10.5 and the forcing $\mathbb{P}_A$ of Theorem 10.3. In its extension the relative uniformizing graphs are $\Pi^2_n[A, p]$. By Lemma 10.7, Lemma 10.4 removes A without adding a parameter or a class alternation. Thus every graph is $\Pi^2_n(p)$, with the same original real parameters, for every finite $n \geq 1$.

The forcing preserves all cardinals of $\mathcal{W}$ and adds no reals. It preserves CH and all projective payoffs and strategy codes, hence preserves PD. This proves the theorem. □

10.5 The two uniformization patterns

Proposition 10.8. *Both L and $\mathcal{W} = L(\mathbb{R})^{M_\omega}$ satisfy Σ^2_n-UP with the same real parameters for every finite $n \geq 1$.*

Proof. First work in $\mathcal{W} = L[A]$ from Lemma 10.5, with A named, and put $H = \mathrm{HC}^{\mathcal{W}}$ and $\kappa = \omega_1^{\mathcal{W}}$. Every $X \subseteq H$ belongs to

$$H^{\mathcal{W}}_{\kappa^+} = L_{\kappa^+}[A].$$

The canonical stage-first order $\prec = <_{L[A]}$ on $\mathcal{P}(H)^{\mathcal{W}}$ has initial segments of size at most κ. It is $\Sigma^2_1[A]$-good in the following sense. There is a first-order certificate $G_A(T, J, C)$ over $(H, \in, A)$ such that

$$\exists C\, G_A(T, J, C) \quad \Longleftrightarrow \quad \{J_i : i \in D_J\} = \{S \subseteq H : S \prec T\}, \tag{P5}$$

where $D_J \subseteq H$ is the active index set of the class code J. For every $T \subseteq H$ there are such J, C. The enumeration is arbitrary, with active indices in H.

Indeed, choose a limit $\beta < \kappa^+$ such that $L_\beta[A]$ contains T, H, A, J and the exact $<_{L[A]}$-initial segment of T. The good $\Sigma_1(A)$ definition of the order supplies this initial segment in a further level below κ^+. Code this relative level, its construction and satisfaction relations, and an enumeration of the predecessors of T which are subsets of H, by classes over H. Anchor the actual H and A, and bind the marked t and listed objects to the actual classes T and J_i by their membership traces on the copies of all $u \in H$. Verify exact relative Def stages. Include a witness for the Σ_1 definition

of $\mathrm{Init}_A(t, I)$, the relative version of Init in Section 3.1, and require the enumeration to contain exactly $I \cap \mathcal{P}(H)$. These are first-order tests, as in Lemma 10.6. Well-founded collapse identifies the actual relative level, and upward $\Sigma_1(A)$ absoluteness identifies I with the full canonical initial segment. This proves (P5) and its existence assertion, also for any prescribed enumeration J of the predecessors. Lemma 10.4 applies with T, J as free class variables, so the good-order relation is also parameter-free Σ^2_1 in $\mathcal{W}$. Equivalently, use the canonical coding of H by κ to obtain the good order on $\mathcal{P}(\omega_1)^{\mathcal{W}}$.

Let $\mathcal{E}(p, x, y)$ be $\Sigma^2_n[A, p]$. Combine the leading existential class witness with y into one class T, using fixed recursive tags. Write

$$\mathcal{E}(p, x, y) \iff \exists T\, [\mathrm{out}(T) = y \wedge \psi_*(p, x, T)],$$

where ψ_* includes validity of the pair code, is first order if $n = 1$, and belongs to $\Pi^2_{n-1}[A, p]$ if $n \geq 2$. Choose the $\prec$-least T satisfying $\psi_*(p, x, T)$ and define

$$\begin{aligned} U(p, x, y) \iff \exists T, J, C\, [&G_A(T, J, C) \wedge \mathrm{out}(T) = y \wedge \psi_*(p, x, T) \\ &\wedge \forall i \in D_J\ \neg\psi_*(p, x, J_i)]. \end{aligned} \tag{P6}$$

For $n = 1$ all tests after the displayed existential class block are first order. For $n \geq 2$, the predecessor-failure condition is $\Sigma^2_{n-1}[A, p]$: its index quantifier is first order, and Choice collects the witnesses over D_J into one class. Its conjunction with $\psi_* \in \Pi^2_{n-1}[A, p]$, preceded by the existential class certificate, is $\Sigma^2_n[A, p]$. Thus (P6) is a $\Sigma^2_n[A, p]$ uniformizer.

Lemma 10.7 and Lemma 10.4 remove A without increasing n or adding a parameter. This proves the assertion for $\mathcal{W}$. For L use the same argument with the empty predicate; no parameter elimination is needed. □

Lemma 10.9. *For each finite $n \geq 1$, boldface Σ^2_n uniformization and boldface Π^2_n uniformization cannot both hold in a model of* ZFC.

Proof. There is a universal Π^2_n relation $\mathcal{U}(e, x, y)$, with real e coding the formula and its real parameter. For arbitrary first-order formulas, include their satisfaction class in the last class block: use its first-order Tarski clauses with a conjunction for an existential block and an implication for a universal block. The satisfaction class is unique and exists, so this introduces no alternation.

Assume Π^2_n uniformization and let $\mathcal{F}(e, x, y)$ be a Π^2_n uniformizer of $\mathcal{U}$. If $\mathcal{U}_e$ is already a function graph, then $\mathcal{F}_e = \mathcal{U}_e$. Now

$$\mathcal{D}(e, y) \iff \neg\mathcal{F}(e, e, y)$$

is Σ^2_n and every section is nonempty, since $\mathcal{F}(e, e, \cdot)$ has at most one value. If Σ^2_n uniformization also held, $\mathcal{D}$ would have a total uniformizer with graph

$\mathcal{K} \in \mathbf{\Sigma}^2_n$. Totality gives

$$\neg\mathcal{K}(e,y) \iff \exists z\,[\mathcal{K}(e,z) \wedge z \neq y],$$

so $\mathcal{K}$ is also $\mathbf{\Pi}^2_n$. Choose e_0 with $\mathcal{U}_{e_0} = \mathcal{K}$. Then $\mathcal{F}_{e_0} = \mathcal{K}$, whereas the chosen value y at e_0 satisfies both $\mathcal{K}(e_0, y)$ and $\neg\mathcal{F}(e_0, e_0, y)$, a contradiction. □

Proof of Corollary 1.3. Proposition 10.8 gives the positive Σ assertions in the two ground models, and Theorems 1.1 and 1.2 give the positive Π assertions in their extensions. Lemma 10.9 gives every displayed failure. The forcings add no reals and preserve ω_1, hence preserve HC and all projective truth. □

In the extensions the ground order still orders the old subsets of HC. New subsets of HC are not in $L[A]$, so this order does not exhaust the enlarged class domain. The ground-model good-order argument therefore does not preserve the Σ pattern under the forcing.

Use of generative AI. I used ChatGPT Astra to help develop the forcing construction and to draft parts of the proofs. The main idea of this work, the deletion strategy is an old idea of mine, actually developed in another context (where it failed to work). The second order context puts enough room between what we code and how the reals behave so that it can be put to work. I also used ChatGPT to improve the language, revise the exposition, avoid typos, and to locate references and relevant literature, in particular the paper [2] which unlocked a much neater proof. I take responsibility for the contents of this paper.

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TU Wien, Institut für Diskrete Mathematik und Geometrie
`stefan.alexander.hoffelner@gmail.com`